

GRADE 11 BIOLOGY

4-DAY EXAM CRAM

Sec.1 Cell Biology → Sec.6 Annelida · Appendices A–D

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6 Sections

120+ Tables

14 SVG Diagrams

Synonyms

Mnemonics

Boundaries

Full Lifecycle Maps

Quick Recall Anchor (all 6 sections):

CBPPAA = **C**ell Biology → **B**acteria → **P**rotista (6 classes) → **P**lasmodium → **P**aramecium → **A**nnelida.

(Earlier shorthand C–B–P–P–A–A is completed here: six sections, one letter each.)

Table of Contents

Sec.1 Cell Biology — Prokaryotic vs Eukaryotic, Organelles, Membrane Transport, Cell Cycle

Sec.2 Bacteria — Kingdom Monera, Structure, Shape, Gram Stain, Reproduction, Recombination

Sec.3 Kingdom Protista — 6 Classes (Rhizopoda, Mastigophora, Ciliophora, Sporozoa, + Algal Groups)

Sec.4 Plasmodium — Classification, Life Cycle (Human + Anopheles), Malaria Types

Exam Model Answers — Figures + probable 4-mark / 4.8-mark model answers

Appendix D — Rapid-Fire One-Page Dumps (6 pages)

Sec.5 Paramecium — Classification, Structure, Locomotion, Nutrition, Reproduction, Life Cycle

Sec.6 Annelida — Classification, Earthworm (Pheretima) Anatomy, Segmentation, Metameres, Reproduction

Appendix A — Master Synonym Table (all 6 topics, one-by-one)

Appendix B — Master Mnemonic Bank (all 6 topics)

Appendix C — Cross-Section Difference Tables (identify / isolate / distinguish)

4-DAY EXAM PLAN — HIGHEST MARKS FIRST

Day 1 · 180 min

Sec.4 Plasmodium life cycle (90) + Sec.5 Paramecium life cycle (90) — two full life-cycle diagrams + species table

Day 2 · 150 min

Sec.6 Annelida (90) + Sec.3 Protista 6-class table (60) — 3-class earthworm comparison

Day 3 · 150 min

Sec.2 Bacteria (90) + Sec.1 Cell Biology (60) — prokaryote/eukaryote + organelle table + cell cycle

Day 4 + night before · 45 min

Appendix B mnemonics + Appendix C difference tables + re-scribble both life-cycle diagrams from memory

Every night: 15-minute “flash recall” using Appendix B mnemonics.

Master Map — All 6 Topics + Domain-Level Features

PIONEER GRADE 11 BIOLOGY — 6-SECTION MAP WITH DOMAIN-LEVEL FEATURES

GRADE 11 BIO

Sec.1 CELL BIOLOGY

Prokaryote vs Eukaryote · Organelles

Membrane transport · Cell cycle

DOMAIN FEATURES — 70S / 9-granule (prok.) vs 80S / 9+2 (euk.) · 4 membrane types: permeable → selective · cycle G1→S→G2→M, DNA 2C→4C→2C · centriole = animal; big vacuole = plant

Sec.2 BACTERIA

Kingdom Monera · Prokaryote

Murein · 70S · Binary fission

DOMAIN FEATURES — murein wall (Gram + / -), Mycoplasma: none · circular DNA, NO histone · plasmids · shapes CDS: coccus / bacillus / vibrio / spiral · recombination CTC; no true meiosis

Sec.3 PROTISTA

Eukaryotic “residuals” · 6 classes

Protozoa + algal divisions

DOMAIN FEATURES — locomotion P-F-C-N (pseudo / flag / cil / none) · nutrition: holozoic · osmotrophic · mixotrophic · Sarcodina → Mastigophora → Ciliophora → Sporozoa — Plasmodium & Paramecium sit HERE (their Sec.4 / Sec.5 come from here)

Sec.4 PLASMODIUM

Apicomplexa / Sporozoa · Malaria

2-host life cycle: human + Anopheles

DOMAIN FEATURES — spp: falciparum · vivax · ovale · malariae · knowlesi · fevers: tertian 48 h · quartan 72 h · quotidian 24 h · asexual in human, sexual in mosquito · hypnozoite = relapse (vivax / ovale only)

Sec.5 PARAMECIUM

Ciliophora · holozoic · 50–300 µm

Slipper-shaped freshwater ciliate

DOMAIN FEATURES — 2 nuclei: macro = soma, micro = germ · 2 contractile vacuoles · oral groove + cilia · life cycle: fission → conjugation → encystation — conjugation = recombination, NOT more cells

Sec.6 ANNELIDA

Segmented worms · ~16,500 spp.

True coelom · Metamerism

DOMAIN FEATURES — 3 classes P-O-H: Poly · Oligo · Hirudinea · coelomic fluid = hydrostatic skeleton · Pheretima: clitellum 34–37, arches 7–11 · oligo + hiru = hermaphrodite, no larva

CROSS-DOMAIN QUICK FACTS — connect all 6 topics (clear boundaries, no overlap)

- Domain split: **Bacteria = PROKARYOTE** — the other 5 topics = **EUKARYOTE** (Eukaryota).
- Protista = the “residual” kingdom: eukaryotes that are NOT plant / animal / fungus.
- Sec.4 & Sec.5 are SPECIES inside Sec.3 — exam wants both full classifications + full life cycles.
- Annelida metamerism = repeated segment + organ blocks → conceptually links to Sec.1 cell/tissue organisation.
- Body-cavity ladder: acoelomate → pseudocoelomate → **COELOMATE** (annelids = true schizocoel, protostome).
- GOTCHA: 70S ribosome = prokaryote AND mitochondrion/chloroplast — never answer “only prokaryotes”.
- Malaria vector = female **Anopheles ONLY** (never Aedes); falciparum = most dangerous (cerebral malaria).
- Earthworms + leeches = hermaphrodite, no larva; polychaetes = separate sexes + trochophore larva.

EXAM PRIORITY (4 days)

Sec.4 + Sec.5 life cycles = highest marks → Sec.6 Annelida → Sec.3 six classes → Sec.2 Bacteria → Sec.1 Cell Biology → Appendices B + C.

Fig 0: Full-page master map — 6 section cards (each with domain-level features), cross-domain quick facts, exam priority strip. Cards and rows are separated by clear gaps; nothing overlaps.

Sec.1 CELL BIOLOGY — Prokaryotic vs Eukaryotic, Organelles, Membrane, Cell Cycle

1.1 Prokaryotic vs Eukaryotic Cell — Master Table

Feature	Prokaryotic (Bacteria, Archaea)	Eukaryotic (Plant, Animal, Fungi, Protista)	Exam Tag
Nucleus	Naked nucleoid (genophore) — no nuclear membrane	True nucleus with double nuclear membrane + pores	MUST
DNA	Circular, no histones (bacteria); circular + histone-like (archaea)	Linear DNA wrapped around histones → chromosomes	MUST
Ribosomes	70S (30S + 50S)	80S in cytoplasm (40S + 60S); 70S in mito/chloro	MUST
Organelles	No membrane-bound organelles (no mito, no ER, no Golgi, no plastid)	Full suite: mito, ER, Golgi, lysosomes, plastids (plant)	HIGH
Cell Wall	Peptidoglycan (murein) in bacteria; pseudopeptidoglycan in archaea; absent in mycoplasma	Cellulose (plant); chitin (fungus); absent in animal	HIGH
Size	0.2–10 µm (most 1–5 µm)	10–100 µm typical; up to 100 µm (ostrich egg)	MED
Division	Binary fission (amitosis in some eukaryotes)	Mitosis (somatic) / Meiosis (germ)	HIGH
Flagella	9 peripheral granules in T.S. (no 9+2 microtubules)	9+2 microtubule arrangement (9 outer doublets + 2 central singlets)	MUST
Mesosomes	Inward folds of plasma membrane (respiration analogue)	Absent (mitochondria handle respiration)	MED
Plasmids	Present (extrachromosomal circular DNA)	Absent in nucleus; mitochondrial/chloroplast DNA is separate	MED
Vacuoles	No true vacuole; gas vacuoles in aquatic bacteria	Large central vacuole (plant); small/transient (animal); tonoplast membrane	MED
Cyclosis	Absent (static cytosome)	Present in plant cells, protists (Amoeba, Elodea)	MED
Example	E. coli, Mycoplasma, Cyanobacteria	Paramecium, Plasmodium, Pteris (fern cell)	MED

BOUNDARY / LIMIT

Prokaryotes NEVER undergo mitosis or meiosis. *Mycoplasma* has NO cell wall (the only eubacteria without one). Mature mammalian RBCs are enucleate but camel RBCs retain a nucleus. Mesokaryotes (dinoflagellates) have a nuclear membrane BUT no histones — they sit at the prokaryote-eukaryote boundary.

1.2 Organelles — Master Comparison (membrane, structure, function, DNA, ribosome, plant/animal)

Organelle	Membrane	Key Structure	Function	Own DNA?	Own Ribosome?	Plant / Animal
Nucleus	Double (nuclear envelope)	Nuclear pores (300–1000 pores), nucleolus, chromatin (hetero- + eu-)	Stores DNA; controls all cell activity; “master controller”	Yes (DNA + histones)	70S in nucleolus (rRNA synthesis)	Both
Mitochondrion (Chondriosome)	Double	Outer + inner membrane, cristae (infoldings), matrix, oxysomes (F1 particles), 70S ribosomes	Aerobic respiration → ATP (oxidative phosphorylation + TCA cycle)	Yes (circular)	70S	Both (more in animal)
Chloroplast	Double	Grana (thylakoid stacks) = light reaction; stroma = dark reaction (Calvin); quantasome (230–250 Chl molecules)	Photosynthesis: solar energy → ATP + NADPH → CO ₂ fixation → glucose	Yes (circular)	70S	Plant only
RER (Rough ER)	Single (cisternae)	Studded with ribosomes on outer surface	Protein synthesis + initial folding; transport pipeline	No	Bound ribosomes	Both
SER (Smooth ER)	Single (tubules)	No ribosomes	Lipid synthesis, glycogen metabolism, detoxification	No	None	Both (more in liver)
Golgi apparatus (Dictyosome in plants)	Single (flattened cisternae)	Cis face (receiving) → Trans face (shipping)	Modifies, packages, and ships proteins; glycosylation; lysosome formation	No	None	Both
Lysosome	Single	0.2–0.8 µm; ~40 acid hydrolases; pH ~5	Intracellular digestion (heterophagy) + autophagy (self-digestion); 4-stage lifecycle	No	No	Both (abundant in animal; rare in plant)

Organelle	Membrane	Key Structure	Function	Own DNA?	Own Ribosome?	Plant / Animal
Peroxisome / Microbody	Single	0.2–1.5 µm; catalase + urate oxidase crystalline core; glyoxysome in plants	Breaks down $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$ (catalase); β -oxidation of fatty acids; photorespiration (glyoxysome)	No	No	Both (glyoxysome = plant only)
Ribosome (Palade's granule)	None (non-membranous)	23 nm; 70S (prokaryote/mito/chloro) or 80S (euk. cytoplasm); polyribosomes (polysomes) on mRNA	Protein synthesis (translation)	No	–	Both
Cytoplasm	–	Cytosol + organelles; ergastic inclusions	Metabolic engine: (reserve food, secretory, excretory); RNA-rich glycolysis, PPP, active exchange	No	None	Both
Vacuole (Tonoplast)	Single (tonoplast = same as plasma membrane)	Central vacuole in mature plant cell (occupies 80–90% of volume); cell-sap inside	Osmoregulation, turgor pressure, storage (minerals, sugars, anthocyanins, waste)	No	No	Plant (large); Animal (small/transient)
Centriole	–	Two perpendicular cylindrical structures (9 triplets of microtubules each)	Organizes mitotic spindle; cilia/flagella base (basal body)	No	None	Animal only (absent in most plants; present in lower plants)
Cell Wall	None (dead structure)	Cellulose (plant); chitin (fungus); peptidoglycan (bacteria); absent in animal	Rigid protection + shape; primary wall (flexible) + secondary wall (rigid, lignified)	No	None	Plant, Fungus, Bacteria
Plastid (Schimper's classification)	Double	Leucoplast (amyloplast, elaioplast, aleuronoplast); Chromoplast (phaeoplast, rhodoplast); Chloroplast	Pigment storage + photosynthesis; interconvertible (but mature leucoplasts in specialized cells NEVER revert)	Yes	70S	Plant only

ORGANELLE MNEMONIC — “N-M-G-L-R-C-V-C-P”

Nucleus · Mitochondria · Golgi · Lysosome · Ribosome · Centriole · Vacuole · Chloroplast · Plastid (Schimper)

“Never Make Green Lizards Run Cold Vibrating Caged Pigeons” — each letter = one organelle to describe in 1-line exam answers.

1.3 Membrane Transport — Permeability Spectrum

Membrane Type	What Passes	What Blocked	Example	Key Detail
Permeable	Solvent + solute (both freely)	Nothing	Cell wall (cellulose pores)	Both liquid and solid particles pass
Impermeable	Nothing	Everything	Polythene bag, plastic sheet	No transport at all
Semipermeable	Solvent only (water)	Solute (all)	Parchment paper, filter paper	Only liquid molecules pass; solutes blocked
Selective / Differentially Permeable	Solvent + specific solutes (1–15 µm)	Solutes > 15 µm, charged ions (passively)	Plasma membrane (all living cells)	Active transport (ATP-driven pumps) can move against gradient; passive = diffusion + osmosis

1.4 Cell Cycle — G1 → S → G2 → M (Mitosis)

Phase	Duration	Key Events	Chromosomes	Chromatids	DNA (C value)	Checkpoint
G1 (Gap 1)	Hours to years (variable)	Cell grows, makes proteins, checks for damage; G0 exit possible	46 (human)	46	2C	G1 checkpoint: size, nutrients, DNA damage
S (Synthesis)	~6 h in human cells	DNA replication: each chromosome → 2 sister chromatids; centrosome duplicates	46	92	4C	Intra-S checkpoint: replication errors
G2 (Gap 2)	~4–6 h	Final growth; spindle proteins made; centrosomes migrate to poles	46	92	4C	G2/M checkpoint: “ready to divide?”
M (Mitosis)	~1 h	Prophase → Metaphase (SAC checkpoint) → Anaphase → Telophase → Cytokinesis	46 → 92 (at anaphase) → 46	92 → 46	4C → 2C each daughter	SAC at metaphase: all kinetochores attached

CELL CYCLE MNEMONIC

“G1 = Grow · S = Synthesize (DNA) · G2 = Get ready · M = Mitosis (divide)” — G-S-G-M = “G-S-G-M: Get Stuff Going, Make.”

BOUNDARY

G0 = permanent exit from cell cycle (mature neurons, cardiomyocytes, muscle fibers). Mammalian RBCs are enucleate → cannot divide by ANY method. Prokaryotes do NOT undergo mitosis — they divide by binary fission only.

Sec.2 BACTERIA — Kingdom Monera (Prokaryotes)

2.1 What is a Bacterium? — Core Identity Card

Item	Detail
Kingdom	Monera (prokaryotic; includes Bacteria + Cyanobacteria + Archaea)
Cell type	Prokaryotic — no true nucleus, no membrane-bound organelles
Nucleoid	Genophore / incipient nucleus — circular DNA + RNA + protein; NO nuclear membrane, nucleoplasm, nucleolus, or chromatin; NO histones
Size	0.2–10 µm (most 1–5 µm); smallest living = <i>Mycoplasma</i> (~0.2 µm)
Distribution	Ubiquitous (everywhere: soil, water, air, extremes, inside other organisms)
Nutrition	Autotrophic (photo/chemo) OR heterotrophic; act as decomposers & mineralisers of the biosphere
Examples	<i>Vibrio</i> , <i>Diplococcus</i> , <i>Bacillus</i> , <i>Streptococcus</i>

2.2 Bacterial Cell Structure — Labelled Diagram

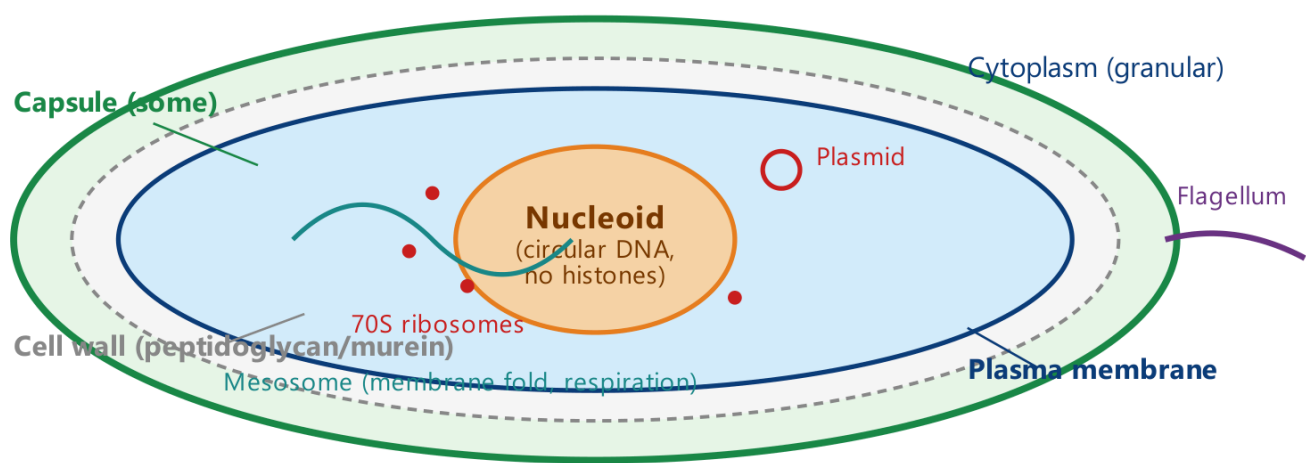


Fig 1: Bacterial cell — outer to inner: capsule → cell wall (murein) → plasma membrane → cytoplasm → nucleoid + plasmid + 70S ribosomes + mesosomes + flagella

2.3 Cell Wall Composition (Critical Boundary Marker)

Wall Type	Composition	Gram Reaction	Key Feature / Limit
Gram-positive	THICK peptidoglycan (20–80 nm) + teichoic acids	BLUE (retains crystal violet)	Sensitive to penicillin (wall target)
Gram-negative	THIN peptidoglycan (5–10 nm) + OUTER LPS membrane	RED/PINK (takes safranin)	LPS = endotoxin; harder for antibiotics to penetrate
Acid-fast (Mycobacteria)	Waxy mycolic acid layer	ACID-FAST (Ziehl-Neelsen)	Resistant to Gram stain + many drugs
No wall (Mycoplasma)	ABSENT	–	Pleomorphic; smallest bacteria; penicillin-RESISTANT

2.4 Cohn's Shape Classification (Morphology) — With Arrangement Terms

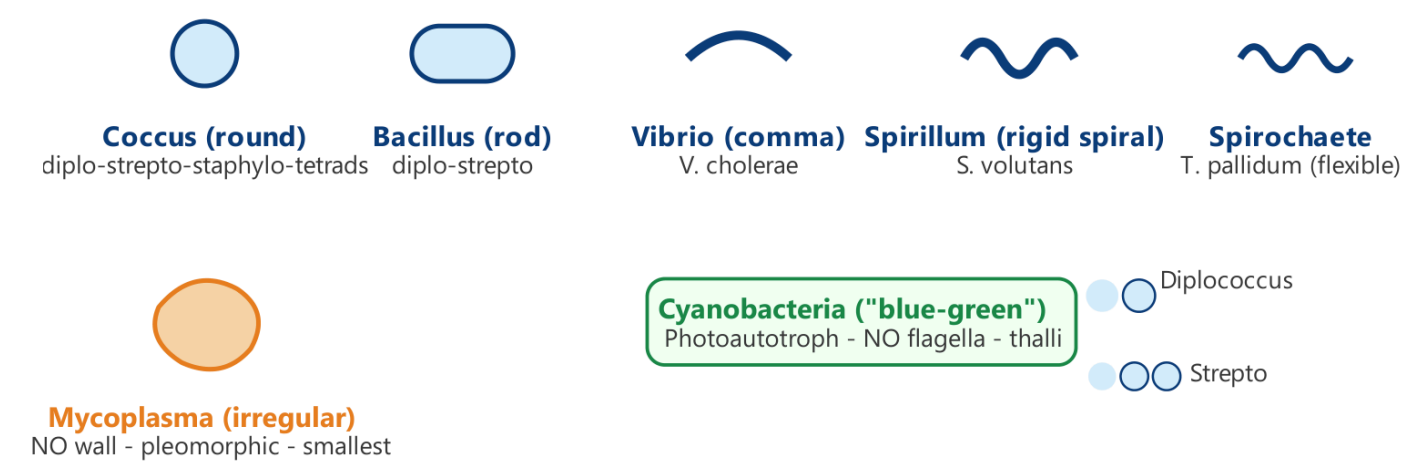


Fig 2: Cohn's shape system — cocci, bacilli, vibrio, spirillum, spirochaete, mycoplasma + arrangement terms + cyanobacteria

2.5 Reproduction & Genetic Recombination

Method	Type	Mechanism	Requirement	Outcome
Binary Fission	Asexual (primary)	DNA replicates → cell elongates → septum forms → 2 identical cells	Nutrients + energy	Genetic clones; exponential (1→2→4→8)
Endospore (sporulation)	Survival	Cell dehydrates → keratin coat → dormant spore (heat/acid/desiccation resistant)	Stress (heat, drought, chemicals)	<i>Bacillus, Clostridium</i> ; NO new cells — germinates when conditions improve
Conjugation	Recombination (sexual)	F ⁺ donor connects to F ⁻ via sex pilus → F plasmid transfers	Cell-to-cell contact	Recipient gains new genes (e.g. antibiotic resistance)
Transformation	Recombination (sexual)	Cell takes up FREE DNA from environment → integrates into chromosome	Free DNA + competent cell	New traits (Griffith's <i>S. pneumoniae</i>)
Transduction	Recombination (sexual)	Bacteriophage carries donor DNA → injects into new host	Phage + two compatible bacteria	Gene transfer via viral vector

2.6 Bacteria vs Archaea (Distinction Card)

Feature	Bacteria	Archaea
Cell wall	Peptidoglycan (murein)	Pseudopeptidoglycan (NO murein)
Membrane lipids	Fatty acids (ESTER bonds)	Isoprenoids (ETHER bonds)
Methanogens?	No	Yes (methane producers)
Extremophiles?	Rare	Common (thermo-, halo-, acidophiles)
Antibiotic sensitivity	Yes (wall target)	Generally resistant

BACTERIA MNEMONICS

- Recombination = "CTC" → Conjugation (cell-to-cell), Transformation (takes DNA), Transduction (transit of phage).
- Shape arrangement = "C-D-S-T" → Coccus → Diplo- → Strepto- → Staphylo- → Tetrads.
- Wall memory = "G-P thick, G-N thin + outer" (Gram-Positive thick, Gram-Negative thin + outer LPS).

BOUNDARY / LIMIT

Bacteria are PROKARYOTES — no nucleus, no mito, no Golgi, no meiosis. 70S ribosomes (30S+50S, NOT additive). Murein is the DEFINITIVE bacterial marker; its absence = mycoplasma. Binary fission is asexual; sexual recombination ONLY via conjugation/transformation/transduction (never syngamy or meiosis). Archaea differ: no murein, ether lipids, histone-like proteins.

Sec.3 KINGDOM PROTISTA — 6 Classes (Full Classification + Locomotion + Nutrition)

3.0 Kingdom Protista — Identity Card

Item	Detail
Domain	Eukaryota
Kingdom	Protista (or Protozoa as the animal-like group; Haeckel 1869 coined "Protista"; de Vienne/Lwoff later split into protozoa/algae/fungi-like)
Nature	Unicellular / colonial / simple multicellular EUKARYOTES; "acellular" (body = one protoplasmic mass); MOSTLY microscopic; free-living or parasitic
Cell wall	Absent in protozoa (flexible pellicle / pellicle + cilia); present in algae (cellulose / siliceous shell in diatoms)
Locomotion	Pseudopodia → Flagella → Cilia → None (passive)
Nutrition	Holozoic (ingestion) → Osmotrophic → Photosynthetic (autotrophs) → Mixotrophic
Reproduction	Asexual (binary multiple fission) + sexual (conjugation, syngamy) + some spore formation

3.1 The 6 Classes — Master Comparison Table (the core exam table)

Class	a.k.a. / synonym	Locomotion	Characteristics	Examples (names only)	Lifestyle
1. Sarcodina (Rhizopoda / Amoeboid)	Rhizopoda · Amoebida · Sarcomastigophora-amoebozoans	Pseudopodia (temporary cytoplasmic projections) — "false feet"	Irregular shape; single or multiple nuclei; no rigid covering; engulf food (phagocytosis); many form cysts (resting spores)	<i>Amoeba (a. proteus)</i> , <i>Entamoeba histolytica</i> , <i>Arcella</i> , <i>Polytoma</i> , <i>Plasmodium</i> (vegetative)	Free + parasitic
2. Mastigophora (Flagellata)	Flagellata · Mastigophora · Euglenophyta · Zoomastigina	Flagella (whip-like, many) — locomotion	Whip-like flagella; some have chloroplasts (euglenoids — mixotrophic, "half-plant half-animal"); pellicle flexibility	<i>Euglena</i> , <i>Trypanosoma</i> (bloodsucker), <i>Giardia</i> , <i>Leishmania</i> , <i>Trichomonas</i> , Dinoflagellates	Free + parasitic
3. Ciliophora (Ciliata / Infusoria)	Ciliata · Infusoria · Oligohymenophorea · Ciliophora	Cilia (short, numerous, rhythmic, cover body)	Body covered with hair-like cilia; MACRONUCLEUS + MICRONUCLEUS (di-primary nucleus — unique!); most complex protozoa; conjugation	<i>Paramecium (caudatum)</i> , <i>Opalina</i> , <i>Stentor</i> , <i>Balantidium/Entamoeba bala</i> , <i>Vorticella</i>	Mostly free, some parasitic
4. Sporozoa (Apicomplexa)	Sporozoa · Apicomplexa · Protozoa (parasitic) · Coccidia	NONE (passive; no locomotion in adults)	Obligate PARASITES; spore-forming; reduced organelles; sexual + asexual cycles (digenetic); many have host alternation	<i>Plasmodium</i> (falciparum, vivax, malariae, ovale), <i>Toxoplasma gondii</i> , <i>Cryptosporidium</i> , <i>Isospora</i>	Parasitic only
5. Chrysophyta / Haptophyta (Golden-brown / Algal)	Golden algae · Haptophyta · Prymnesiophyta	Some with flagella (one tinsel-type)	Golden-brown (fucoxanthin + chlorophyll a,c); unicellular or colonial; plant-like; photosynthetic; some have calcium carbonate statoliths	<i>Chrysmpha</i> , <i>Prymnesium</i> , diatoms (<i>Diatoma</i> — siliceous test), Chrysophyceae	Free (aquatic)
6. Cyanophyta / "Euglenophyta / Algae-like" (Blue-green → now Bacteria / or euglenoids)	Cyanophyta · Cyanobacteria ("blue-green algae") · Euglenophyta	NONE (cyanobacteria); flagellum (euglenoids)	Cyanobacteria: photosynthetic PROKARYOTES, no true nucleus, 70S, N-acetylmuramic acid cell wall, store cyanophycean starch, heterocysts fix N ₂ — Euglena: mixotrophic protist (photo + chemo), red "eyespot" (stigma), flexible pellicle	<i>Nostoc</i> , <i>Anabaena</i> (heterocyst), <i>Oscillatoria</i> , <i>Spirulina</i> ; <i>Euglena (v. gracilis)</i>	Free (aquatic / soil)

BOUNDARY — THE "6 CLASSES" EXAM TRAP

Different textbooks number the "6 classes of Protista" differently. The Pioneer/NEB convention typically lists: Sarcodina (Rhizopoda) – Mastigophora (Flagellata) – Ciliophora – Sporozoa – + the ALGAL divisions. Your book (Gurung & Ghimire) most likely splits the algal portion into two divisions (e.g., Chrysophyta + Cyanophyta, or Dinoflagellata + Euglenophyta). Action: check YOUR book's exact 6 list on the Protista page and write down its 6 names verbatim before the exam. The 4 ANIMAL-LIKE (protozoan) classes are Sarcodina, Mastigophora, Ciliophora, Sporozoa — these are the ones with classifications and life-cycles (Sec.4, Sec.5 come from here).

3.2 Protozoan Locomotion — 4 Types (Identify / Distinguish)

Structure	How it moves	Class using it	Distinguishing feature
Pseudopodia	Amoeboid streaming of cytoplasm; "false feet" extend & flow	Sarcodina	Irregular, slow, direction can change; no set rhythm

Structure	How it moves	Class using it	Distinguishing feature
Flagella	Whip-like undulation of 1 or many long filaments	Mastigophora	Fast; 1 (e.g. <i>Trypanosoma</i>) to many (<i>Giardia</i>)
Cilia	Rhythmic coordinated beating of many short hairs over the body	Ciliophora	Reversible; "ciliary current" drives food to cytopharynx; many = fastest protozoan
None (passive)	Drift / host blood flow; motile forms (sporozoites) glide	Sporozoa	Adults non-motile; gametophytic forms use host movement

3.3 Nutrition Types in Protista

Type	Meaning	Example
Holozoic (ingestion)	Engulfs particles, digests intracellularly (food vacuole + lysosome)	<i>Paramecium</i> , <i>Amoeba</i>
Osmotrophic / absorptive	Absorbs dissolved food through surface	Fungi-like protists, slime moulds (<i>Fuligo</i>)
Photosynthetic (holophytic)	Self-feeding via chloroplasts	Algae, <i>Euglena</i> (light phase)
Mixotrophic	Switches between photosynthesis AND heterotrophy	<i>Euglena</i> (dark → osmotrophic; light → photosynthetic)

PROTISTA 4-ANIMAL-CLASS MNEMONIC — "S-M-C-S"

Sarcodina (Sneaky pseudopod) · Mastigophora (Masts = flagella) · Ciliophora (Covered in cilia, two nuclei) · Sporozoa (Still/parasite, spores).

"Some Mosquitoes Catch Sporozoa" — the 4 protozoan classes; add your book's 2 algal divisions for the full 6.

Sec.4 PLASMODIUM — Complete Classification + Full Life Cycle (Malaria)

4.1 Complete Classification (exam-critical)

Rank	Name	Note
Domain	Eukaryota (Eukarya)	Eukaryotic cell
Kingdom	Protista (Protozoa)	Animal-like protist
Phylum / Superclass	Protozoa (→ Apicomplexa)	Apicomplexan parasite
Class	Sporozoa (Sporozoans)	Spore-forming, no motile cilia/flagella in adult
Subclass / Subphylum	Apicomplexa (Cnidospora)	Apical complex (polar rings, rhoptries)
Order	Eucoccidiorida (Eucoccidina / Haematozoa)	–
Family	Plasmodiidae (Haemoprota)	–
Genus	Plasmodium	–
Species (5 human pathogens)	<i>P. falciparum</i> · <i>P. vivax</i> · <i>P. ovale</i> · <i>P. malariae</i> · <i>P. knowlesi</i>	NCBI lineage: Eukaryota → Sar → Alveolata → Apicomplexa → Aconoidasida → Haemosporida → Plasmodiidae → Plasmodium

BOUNDARY

Plasmodium is a digenetic (two-host) obligate intracellular parasite. It completes its life between TWO hosts: human (definitive host) + female *Anopheles* mosquito (intermediate host) with host alternation. The asexual (sporogonic = erythrocytic + hepatic) cycle is in the human; the sexual cycle (gametogenesis) is in the mosquito. Vector = female **Anopheles** (only *Anopheles* carries it; *Aedes* does NOT).

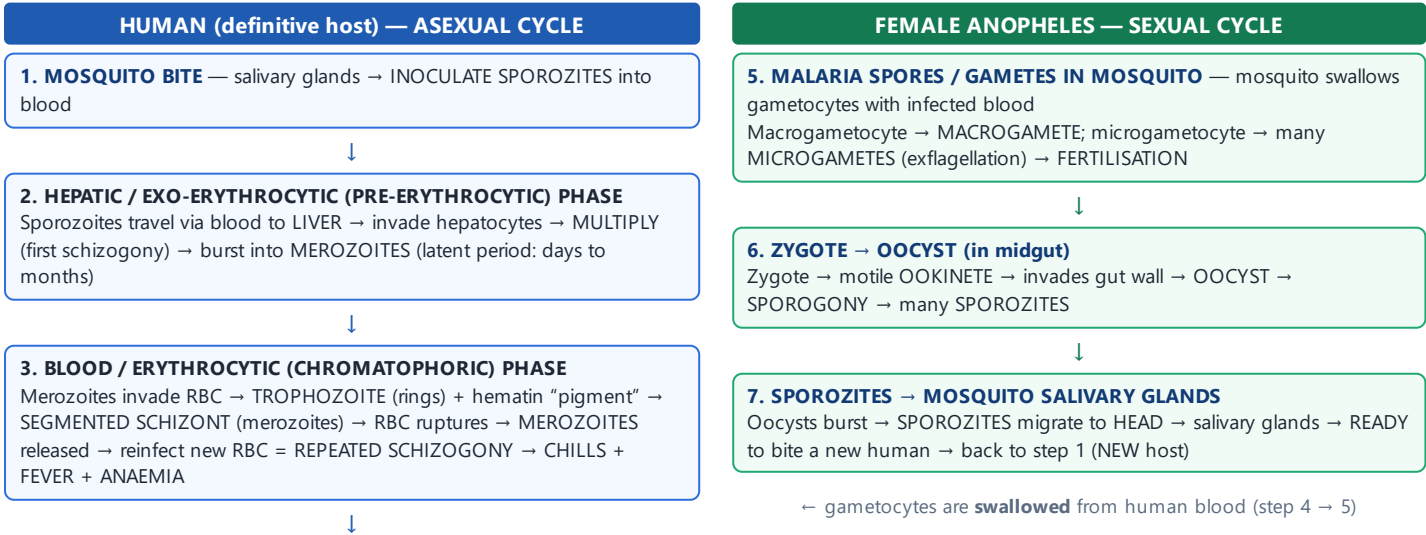
4.2 General Characteristics

- Unicellular, eukaryotic, non-motile adult (no flagella/cilia/pseudopodia); body = membrane + cytoplasm + nucleus
- Obligate parasitic; lives in human liver cells (exoerythrocytic) + RBCs (erythrocytic) + mosquito gut (sexual)
- No food vacuole, no contractile vacuole in adult forms; ingests RBC hemoglobin
- Reproduction: asexual (multiple fission / schizogony / sporogony) + sexual (gametogony) with alternation of generations
- Causes **MALARIA**; periodic chills + fever (“tertian” / “quotidian” / “quartan”)

4.3 The 4 Forms (Morphology) — Identify Each

Form	Shape	Where	Role
Trophozoite (young form)	Amoeboid, growing	RBC / liver	Grows + ingests food; “feeding” stage
Segmented schizont / Merozoite (mature asexual)	Divided into many daughter nuclei (rosette / daisy in vivax)	RBC / liver	Rupture → releases merozoites → next infection
Macrogametocyte (female gametocyte)	Round / kidney	Human blood	Sexual; ingested by mosquito → egg (oocyst)
Microgametocyte (male gametocyte)	Crescent / banana (esp. falciparum)	Human blood	Sexual; produces many motile microgametes in mosquito

4.4 COMPLETE LIFE CYCLE — Full Diagram (the mark-winner)



4. GAMETOGENESIS (sexual begins in blood)
 Some trophozoites → MACROGAMETOCYTES (♀) + MICROGAMETOCYTES (♂) in the RBC (infective forms)

LATENT PHASE (P. vivax & P. ovale ONLY)
 Hepatic schizonts form HYPNOZOITES (dormant liver stages) → reactivate months later = RELAPSE (not in falciparum/malariae)

Asexual (multiple fission) = **HUMAN** · Sexual (gametogony) = **MOSQUITO**

Fig 3: Plasmodium complete life cycle — 7 numbered stages spanning HUMAN (asexual: hepatic + blood schizogony) and FEMALE ANOPHELES (sexual: gametogony + sporogony), plus the vivax/ovale hypnozoite relapse box

4.5 Malaria Types (species → fever pattern) — Distinguish Table

Species	Fever pattern	Key Distinguishing Feature	Severe?
P. falciparum	Irregular / continuous	Most dangerous (cerebral malaria); crescent microgametocytes; bands of trophozoites at RBC edge; NO Schuffner's dots	YES
P. vivax	Tertian (48 h)	INVades OXYGENATED RBCs (not just young); HYPNOZOITES → relapse; Schuffner's dots present	Moderate
P. ovale	Tertian (48 h)	INVades YOUNG (reticulocyte) RBCs; Schuffner's dots; RBC slightly smaller + pointed; hypnozoites → relapse	Moderate
P. malariae	Quartan (72 h)	"Rosette" / daisy-head schizont; banana-shaped pigments; mild; chronic	Low
P. knowlesi	~24 h (quotidian)	Zoonotic (macaque → human); fast replication	Possible

MALARIA FEVER MNEMONIC — "QTTK"

Quartan (72 h) = *malariae* · Tertian (48 h) = *vivax* & *ovale* · (falciparum = irregular) · **K**nowlesi = 24 h. "Quoted Tertian twins are Know".

Sec.5 PARAMECIUM (caudatum) — Full Classification, Structure & Complete Life Cycle

5.1 Complete Classification (exam-critical)

Rank	Name	Note
Domain	Eukaryota	Eukaryote
Kingdom	Protista (Protozoa)	Animal-like
Phylum	Ciliophora	Ciliated protozoa
Subphylum	Intramacronucleata	–
Class	Oligohymenophorea	–
Subclass	Peniculidia / Peritrichia	–
Order	Peniculida	–
Family	Parameciidae	–
Genus	Paramecium	Ehrenberg 1833/34
Species	caudatum	“cauda” = tail (posterior pointy)

5.2 General Characteristics

- Unicellular, fresh-water, free-living ciliate; lives in ponds, ditches, stagnant water; ~50–300 µm; slipper-shaped (shoe-pan shape)
- Body covered with uniform rows of CILIA (locomotion + feeding current)
- DIPrimary nucleus (unique): MACRONUCLEUS (somatic, “body” functions) + MICRONUCLEUS (reproductive, “germ” functions)
- Organelles: food vacuoles, contractile (pulsatile) vacuoles (2, one each end — osmoregulation), oral groove + cytopharynx (mouth)
- Nutrition: **holozoic** (ciliary current sweeps food to oral groove → food vacuole → lysosomal digestion)
- Excretion: contractile vacuoles (water); excretory vacuoles; waste → cytocyst / anus
- Respiration + excretion: by DIFFUSION through pellicle/plasma membrane (no special organ)

5.3 Structure — Labelled Diagram

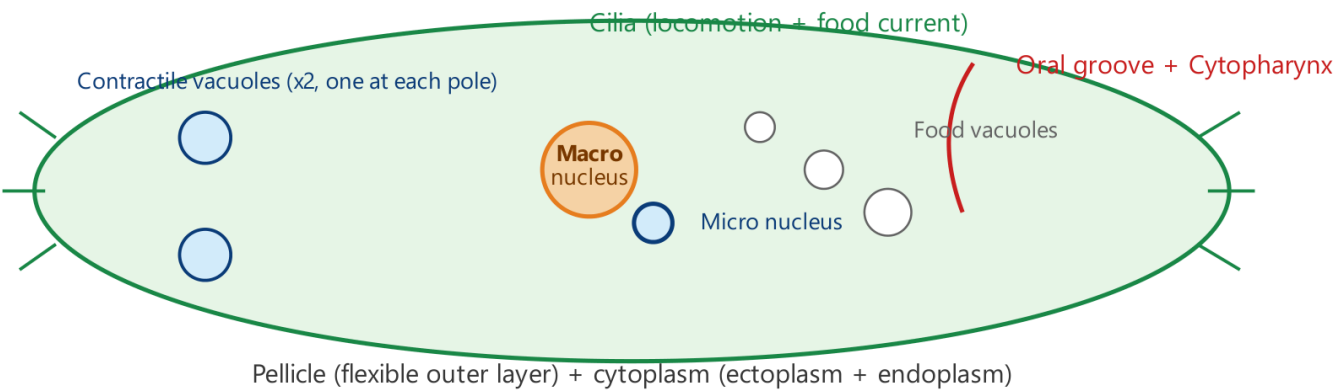


Fig 4: Paramecium caudatum — slipper shape, cilia all over, oral groove (right/anterior), 2 nuclei, food vacuoles, 2 contractile vacuoles

5.4 COMPLETE LIFE CYCLE — 3 Processes

A. ASEXUAL — BINARY FISSION (vegetative)

Both nuclei divide; macro by fission (daughter macros), micro by MEIOSIS → 4 micros, 3 lost, 1 kept; cytoplasm divides → 2 daughter cells (longitudinal) = rapid population increase; main method in healthy pond [1 cell → 2 → 4 → 8 ...] (each division halves size then grows)

B. SEXUAL — CONJUGATION (amalgamation)

Two cells (an EXCONUGATE pair) meet, fuse at cytopharynx → Macro nuclei fragment + disappear → each micro undergoes 3 MEIOTIC divisions → 4 haploid micros; 2 lost; 1 CONJUGANT + 1 RESIDUAL → conjugant (haploid) of each cell SWAPS (reciprocal) → forms SYNKARYON (diploid) → cells separate → 2 EXCONJUGANTS, each reconstitutes a new macro (from synkaryon) + micro → new variety = genetic recombination (not more cells; both survive)

C. ENCYSTATION (resting / survival)

In dry / nutrient-poor water, cell retracts cilia → secretes a tough COAT (2–3 layers) → CYST / HYPNOSPORE (metabola) → metabolic activity pauses.
Favours survival of drought → when water returns, coat dissolves → cell emerges (GERMINATION)

D. HYPNOSPORE → RESYNTHESIS (after conjugation)

Exconjugants in unfavourable conditions form HYPNOSPORES; later → resynthesis (new ciliate from cyst) = "resting spore" = temporary survival structure

LIFE-CYCLE SUMMARY (1 diagram, 4 boxes)

Normal pond: binary fission (fast) → population grows · Favourable: 2 cells conjugate (genetic mix) → new exconjugants ·
Unfavourable: encystation / hypnospore (survival) → when water returns, germination → new ciliate.

Fig 5: Paramecium caudatum complete life cycle — 4 boxes: (A) Asexual binary fission, (B) Sexual conjugation (reciprocal, haploid exchange → diploid), (C) Encystation (resting), (D) Hypnospore → resynthesis. Key distinction: conjugation = genetic recombination (NOT more cells); binary fission = multiplication.

PARAMECIUM MNEMONIC

- "DIPrimary nuclei: Macro = somatic, Micro = germ" — MacroM / MicroM = "**Macro-Micro, Management vs Meiosis**"
- Locomotion = **Cilia** (fast, reversible); nutrition = **holozoic** (oral groove); osmoregulation = **2 contractile vacuoles**
- Life cycle 3 = "**F-C-E**" = **F**ission (asexual) · **C**onjugation (sexual) · **E**ncystation (survival)

BOUNDARY

Paramecium NEVER multiplies by conjugation (conjugation makes 2 NEW VARIETIES, not 4 daughters). Micro nucleus divides by MEIOSIS; macro does not. The MICRONUCLEUS = "the germline"; the MACRONUCLEUS = "the body line". Encystation ≠ reproduction — it is a RESTING stage that lets the cell survive drought; it is the reason Paramecium persists in dried pond scrapes.

Sec.6 ANNELIDA — Segmented Worms (Earthworm Pheretima focus)

6.1 Complete Classification

Rank	Name	Note
Domain	Eukaryota	–
Kingdom	Animalia	Metazoa
Phylum	ANNELIDA	~16,500–17,000 spp.; “annelus = small ring” (segmentation)
Subphylum / Clade	Scaphia (Polychaeta), Hirudinea, Crystallaria	3 main classes
Class 1	Polychaeta (Scaphopoda)	Marine; many bristle-bearing parapodia + cirri; free-living; e.g. <i>Nereis</i> (sea horse worm), <i>Sipuncula</i>
Class 2	Oligochaeta	Few-bristle (oligochete) terrestrial/freshwater worms; setae, no parapodia; e.g. <i>Pheretima</i> (earthworm), <i>Lumbricus</i>
Class 3	Hirudinea	Leeches; few/absent setae; suckers (oral + caudal); parasitic/depredatory; e.g. <i>Hirudo</i> , <i>Amphiglossus</i> , <i>Placobdella</i>

6.2 General Characteristics (Phylum-level — memorise these)

- **Body:** Elongated, cylindrical, metameric (segmented) — each segment (metamer / segment) is a functional unit with repeated organs
- **Symmetry:** Bilateral; **Body cavity:** TRUE COELOM (schizocoelous = from mesoderm split) = schizocoelomate
- **Germ layers:** Triploblastic (ecto-meso-endoderm); Protostome (mouth first)
- **Coelom:** true, filled with coelomic fluid; acts as HYDROSTATIC SKELETON → enables burrowing + peristaltic locomotion
- **Exterior:** Cuticle over epidermis; body wall muscles = circular + longitudinal (in annelids: 4 layers of body wall)
- **Respiration:** By body-surface diffusion (terrestrial/freshwater); gills in polychaetes
- **Excretion:** Nephridia (one pair per segment; metanephridia + protonephridia types)
- **Circulation:** Closed (vessels + “heart” aortic arches in earthworm; no heart = 2–3 pairs of arches)
- **Nervous system:** Ventral nerve cord; brain (supra-oesophageal ganglion) + sub-oesophageal ganglion
- **Reproduction:** Hermaphrodite (*Pheretima* & leeches) or dioecious (polychaetes); asexual regeneration (head) in polychaetes

6.3 The 3 Classes — Quick Comparison

Feature	Polychaeta (Scaphia)	Oligochaeta	Hirudinea
Parapodia	Present (fleshy lobes + setae + cirri)	Absent	Absent (suckers instead)
Setae	Many (polychaete = “many bristles”)	Few (oligochete = “few bristles”)	Few / none
Habitat	Marine	Terrestrial / freshwater	Freshwater / marine
Habit	Free-living	Free-living	Parasitic / depredatory
Sex	Dioecious (separate sexes) + larval (trochophore)	Hermaphrodite, no larva	Hermaphrodite, no larva
Example	<i>Nereis</i> (sea horse / sandworm)	<i>Pheretima</i> (earthworm)	<i>Hirudo</i> (leech)
Metamerism	Homomeric (repeat segments)	Homomeric → metamerism (clitellum + genital segments differ)	Fixed 21 segments

6.4 Earthworm Pheretima / Lumbricus — Body Structure

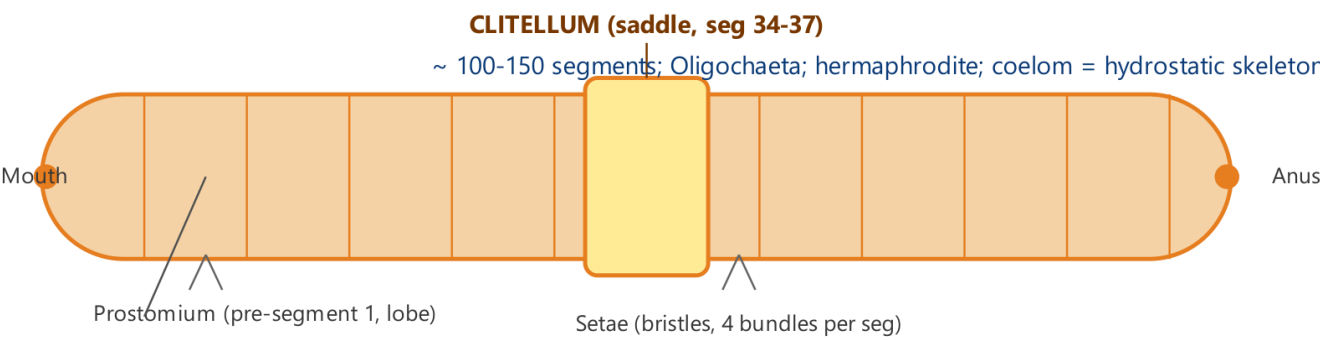


Fig 6: Earthworm (Pheretima / Lumbricus) — segmented body, clitellum (saddle, segments 34-37), setae (bristles), prostomium; internal = coelom + hydrostatic skeleton + metanephridia + ventral nerve cord + aortic arches ("heart")

6.5 Metamerism (Segmentation) — The Exam Concept

Term	Meaning	In Annelida
Metamer	One body segment (a repeated unit)	Earthworm has ~100–150 segments
Metamerism	Repetition of similar body parts in segments	Exterior + interior (organs) both segmented = "double metamerism"
Homomeric metamerism	All segments similar	Polychaetes; most earthworm segments
Heteromeric metamerism	Segments specialised differently	Earthworm: clitellar (34-37) + genital segments differ

6.6 Earthworm Reproduction & Key Internal Systems

System	Key Facts
Reproduction	Hermaphrodite (self-fertilisation IMPOSSIBLE — cross-fertilisation); two males + two female genital pores; CLITELLUM secretes cocoon; eggs + sperm in cocoon; cross-fertilisation; embryo → new worm; NO larva (direct development)
Circulation	Closed; coelomic fluid in vessels; 2–3 pairs of aortic arches ("hearts") in seg 7–11 pump blood; dorsal + ventral vessels
Excretion	Metanephridia (one pair per segment) — flame cup (nephrostome) → tubules → nephridiopore (to coelom then outside)
Nerve	Ventral cord; supra-oesophageal "brain" ganglion + sub-oesophageal ganglion; perivisceral nerves; giant fibres
Locomotion	Peristalsis: circular + longitudinal muscles + coelomic fluid (hydrostatic) + setae anchor; no skeleton
Respiration	Through moist body-surface (diffusion); in polychaetes via parapodia/gills

ANNELIDA MNEMONIC — "3-H-T"

3 classes (Polychaeta, Oligochaeta, Hirudinea) · Hermaphrodite (oligochaeta + hirudinea) · Truly coelomate + T-riploblastic + T-wo nerve centres.
Classes = "P-O-H" → "Polychaeta (many bristles, sea) · Oligochaeta (few bristles, soil) · Hirudinea (sucker, leech)".

BOUNDARY

Annelida = TRUE coelom + metamerism + schizocoelous protostome. Polychaeta have PARAPODIA + separate sexes + trochophore larva; Oligochaeta & Hirudinea = no parapodia, hermaphrodite, no larva. Earthworm's clitellum (seg 34-37) + the aortic arches ("heart" in 7-11) are exam identifiers. Nephridia are the metanephridial type (one per segment).

APPENDIX A — Master Synonym Table (one name = many labels, organised 1/1/1)

Each entry is one concept with its alternate labels — reduces word confusion. Learn them grouped.

Concept	Synonym / Alternate label (1/1/1)
Bacterial cell wall	Peptidoglycan / Murein / Mucopeptide (NAG + NAM)
Bacterial nucleus	Nucleoid / Genophore / incipient nucleus / naked nucleus / prokaryon
Bacterial respiration	Mesosome / mitochondrial-like infolding / plasma-membrane fold
70S ribosome	Prokaryotic ribosome / bacterial ribosome / 30S+50S subunits
Cell division types	Amitosis (direct) / Mitosis (indirect/divisional) / Meiosis (reductional)
Asexual reproduction	Vegetative reproduction / parthenogenesis / clonal / amitosis (in some protists)
Mastigophora	Flagellata / euglenoids / zoomastigina / protomonad flagellates
Sarcodina	Rhizopoda / Amoebida / amoeboid protists / sarcomastigophora-amoebozoans
Ciliophora	Ciliata / Infusoria / Oligohymenophorea / "paramecia"
Sporozoa	Apicomplexa / Coccidia / Protozoa-parasitic / Cnidospora
Plasmodium life stages	Sporozoite (infective, mosquito→liver) / Merozoite (asexual daughter) / Trophozoite (feeding) / Schizont (segmented) / Gametocyte (macrogametocyte + microgametocyte)
Malaria fever cycles	Tertian (48 h: vivax/ovale) / Quartan (72 h: malariae) / Quotidian (24 h: knowlesi) / falciparum = irregular
Paramecium nuclei	Macronucleus (somatic / body / "amitotic") / Micronucleus (reproductive / germ / "meiotic")
Paramecium mouth	Oral groove / cytopharynx / "mouth"
Resting cell	Cyst / Hypnospore / metabola / dormant spore
Earthworm segment	Metamer / segment / body-unit / somite (with coelom) / segment-somite pair
Earthworm "heart"	Aortic arches / arches of Bole / heart-arcs (seg 7-11)
Earthworm excretion	Metanephridia / segmental kidneys / nephridia (one pair per segment)
Earthworm saddle	Clitellum / copulatory gland (seg 34-37)
Hydrostatic skeleton	Coelomic fluid skeleton / fluid-filled body-wall / annelid hydrostatic system

APPENDIX B — Master Mnemonic Bank (all 6 topics, for 4-day cram)

Sec.1 CELL

"N-M-G-L-R-C-V": Nucleus, Mitochondria, Golgi, Lysosome, Ribosome, Centriole, Vacuole, (Chloroplast, Plastid). "Never Make Green Lizards Run Cold Vibrating Caged Pigeons".

Sec.2 BACTERIA

"CTC" + "G-P thick, G-N thin": Conjugation/Transformation/Transduction; Gram-Positive thick wall, Gram-Negative thin + LPS outer. Shapes "C-D-S-T" (Coccus→Diplo→Strepto→Staphylo).

Sec.3 PROTISTA 6

"S-M-C-S + 2 algal": Some Mosquitoes Catch Sporozoa (+ your book's 2 algal divisions). Locomotion "P-F-C-N" (Pseudopodia, Flagella, Cilia, None).

Sec.4 PLASMODIUM

"QTTK" + 7-step cycle: Quartan=malariae, Tertian=vivax/ovale, falciparum=irregular, Knowlesi=24 h. Cycle: BITE → LIVER (merozoites) → BLOOD (schizonts) → GAMETOCYTES → mosquito FERTILISATION → OOCYST → SALIVARY GLANDS → back to bite.

Sec.5 PARAMECIUM

"F-C-E" + "M-M": Fission (asexual), Conjugation (sexual), Encystation (survival); Macro=Management (body), Micro=Mutation/meiosis (germ). 2 contractile vacuoles + oral groove + cilia.

Sec.6 ANNELIDA

"3-H-T" + "P-O-H": 3 classes, Hermaphrodite, Truly coelomate; classes = Polychaeta (parapodia, sea) / Oligochaeta (setae, soil) / Hirudinea (suckers, leech).

APPENDIX C — Cross-Section Difference Tables (identify / isolate / distinguish)

C.1 Protista 4 protozoan classes — one-line distinction

Class	The ONE thing that identifies it
Sarcodina	Pseudopodia ("false feet") — irregular crawling
Mastigophora	Whip-like flagella — fast swimming; some have chloroplasts (<i>Euglena</i>)
Ciliophora	Many short cilia + TWO nuclei (macro+micro) — most complex protozoan
Sporozoa	Obligate parasite, NO motile cilia/flagella, spore-forming (<i>Plasmodium</i> , <i>Toxoplasma</i>)

C.2 Mitosis vs Meiosis vs Amitosis (if asked)

Feature	Amitosis	Mitosis	Meiosis
Divisions	1, no spindle	1, spindle present	2 consecutive (reductional + equational)
Ploidy	$2n \rightarrow 2n$	$2n \rightarrow 2n$	$2n \rightarrow n$
Daughter cells	2 (unequal, unreliable)	2 identical	4 genetically different
Where	Liver, some protists, cancer	All somatic	Germ cells only (gonads)
Variation	None (and errors)	None	Crossing-over + independent assortment

C.3 Prokaryote vs Eukaryote — the 5 must-knows

Point	Prokaryote	Eukaryote
Nucleus	Naked nucleoid	Membrane-bound nucleus
Ribosomes	70S (30S+50S)	80S cytoplasmic; 70S in mito/chloro
DNA	Circular, no histones	Linear, with histones
Flagella T.S.	9 granules (no 9+2)	9+2 microtubules
Organelles	No membrane-bound	Full membrane-bound set

C.4 4-Day Exam Prioritisation Matrix

Priority	Topic	Why	Study Time
1st	Sec.4 Plasmodium full life cycle	Highest-mark single diagram; species/fever table	Day 1: 90 min
2nd	Sec.5 Paramecium full life cycle	Conjugation + encystation + 3 nuclei = marks	Day 1: 90 min
3rd	Sec.6 Annelida	3 classes + earthworm anatomy + metamerism	Day 2: 90 min
4th	Sec.3 Protista 6 classes	Table: class / locomotion / nutrition / example	Day 2: 60 min
5th	Sec.2 Bacteria	Structure + shape + Gram + recombination	Day 3: 90 min
6th	Sec.1 Cell Biology	Organelles + prokaryote/eukaryote + cell cycle	Day 3: 60 min
All	Appendix B mnemonics + C difference tables	Rapid-fire recall before exam	Day 4 + night-before: 45 min

4-DAY EXAM PLAN

- **Day 1 (today):** Sec.4 Plasmodium + Sec.5 Paramecium (two full life-cycle diagrams + species table).
- **Day 2:** Sec.6 Annelida + Sec.3 Protista (6-class table + 3-class earthworm comparison).
- **Day 3:** Sec.2 Bacteria + Sec.1 Cell Biology (prokaryote/eukaryote + organelle table + cell cycle).
- **Day 4 + night before:** Appendix B mnemonics + Appendix C difference tables + re-scribble both life-cycle diagrams from memory.
- **Every night:** 15-min "flash recall" using Appendix B mnemonics.

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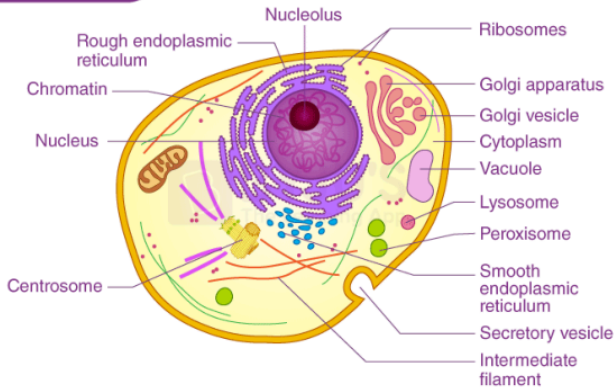
EXAM MODEL ANSWERS + BEST DRAWN FIGURES (Organized by Topic)

Use this as your answer-key. For each topic: (1) plain-language "what this means", (2) the best drawn figure your book tests, (3) similarity / difference / synonym, and (4) a probable 4-mark and 4.8-mark model answer. Draw the figure + write 4 points = 4 marks; add 1 more point + the figure detail = 4.8.

1 — CELL BIOLOGY — figures + probable answers

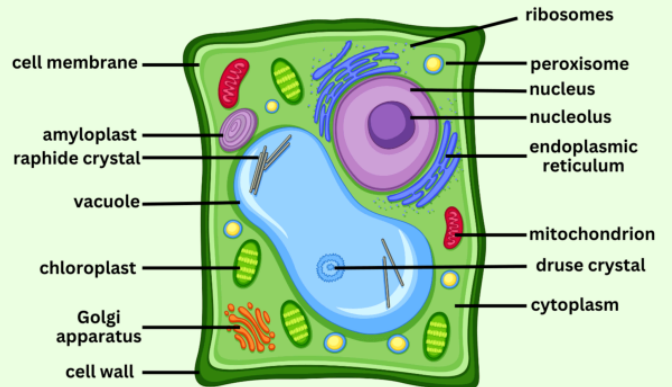
What this means (plain): A cell is the smallest living unit. Prokaryote (bacteria) = no real nucleus, tiny; eukaryote (you, plant, fungus, protist) = true nucleus + organelles. The exam wants you to label the organelles and state one function each.

ANIMAL CELL

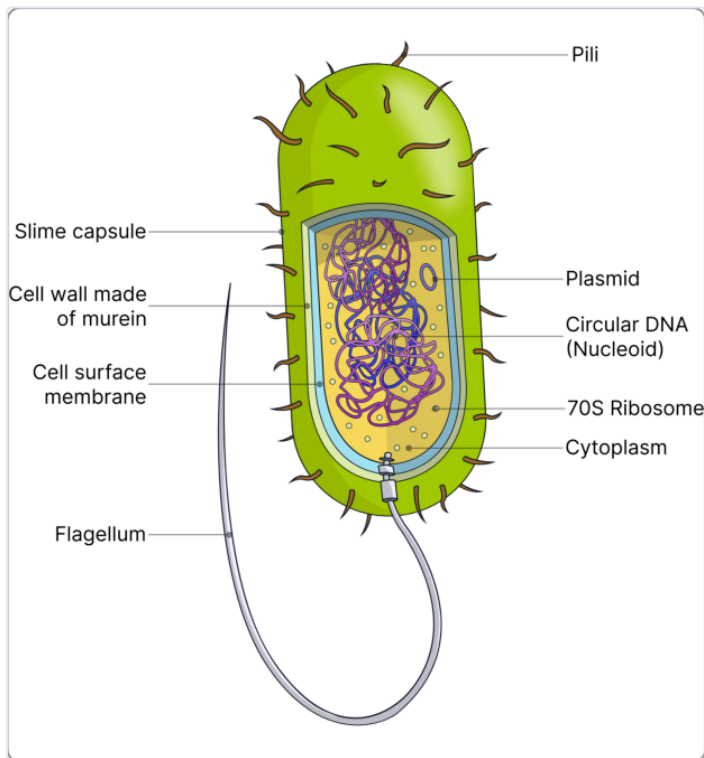


Best drawn figure: Eukaryotic ANIMAL cell (label all 10 organelles)

Plant Cell



Plant cell = animal + cell wall + chloroplast + big vacuole, NO centriole



Prokaryotic (bacterial) cell: nucleoid, murein, flagella, mesosome, 70S

Nucleus

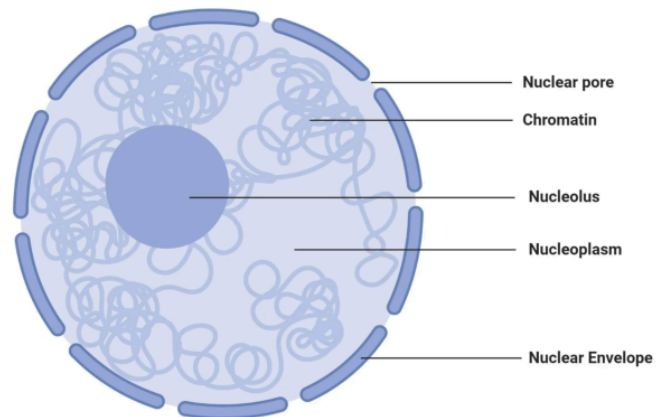
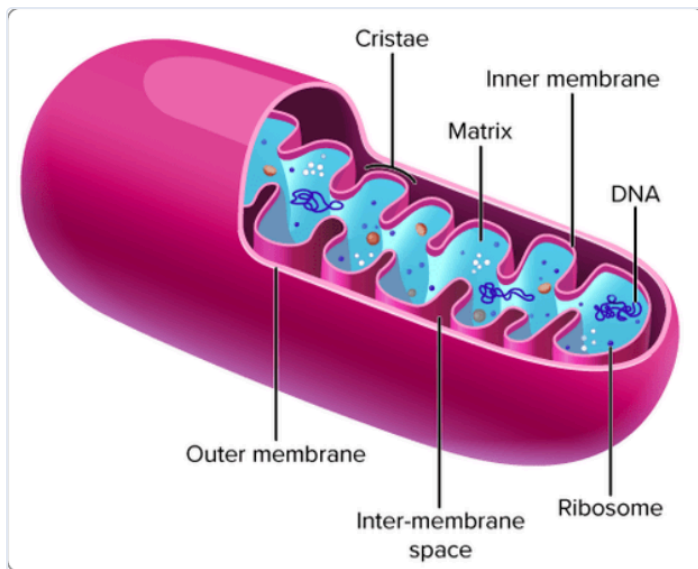
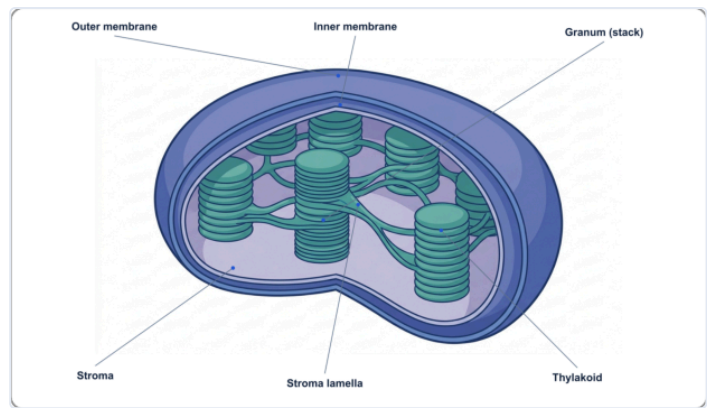


Figure: Nucleus, Image Copyright Sagar Aryal, www.microbenotes.com

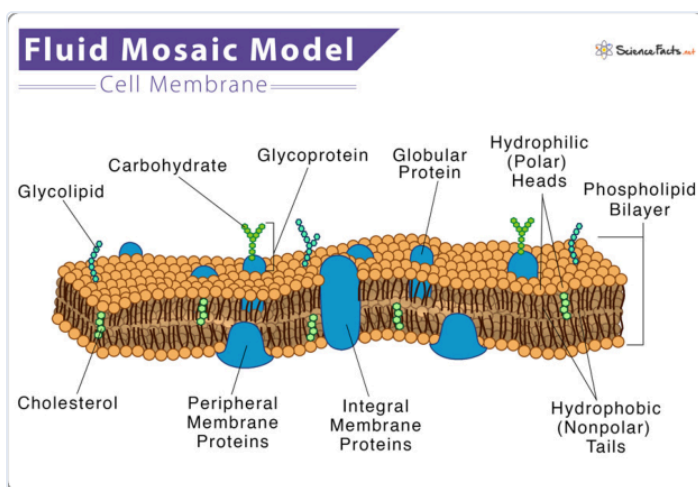
Nucleus: envelope + pores + nucleolus + chromatin (draw & label)



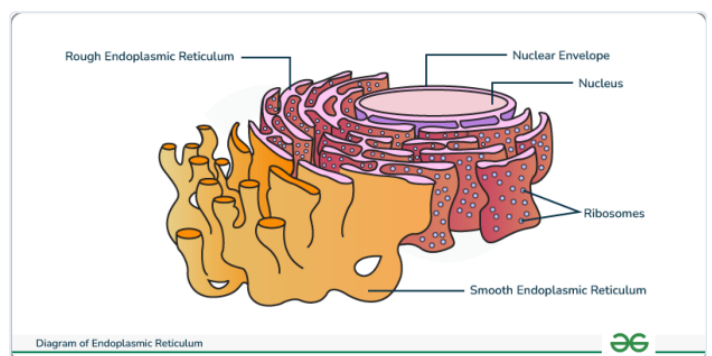
Mitochondrion: outer/inner membrane, cristae, matrix, 70S ribosomes



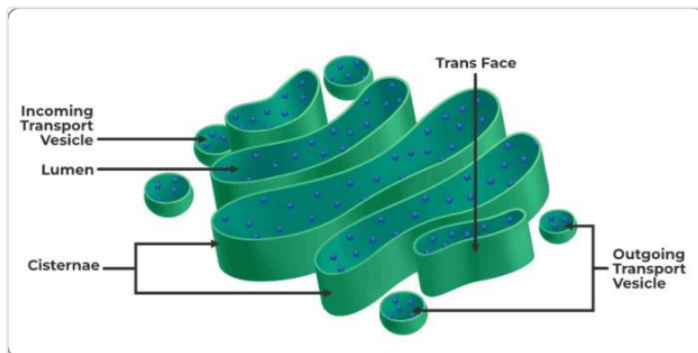
Chloroplast: grana (thylakoid stack) + stroma; light rxn in grana



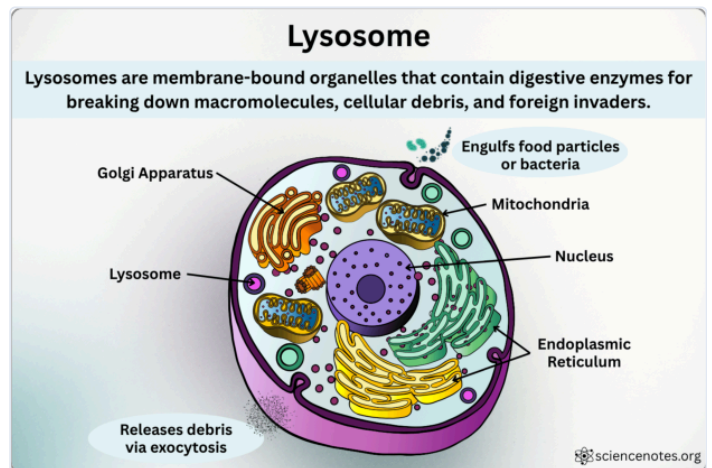
Cell membrane — FLUID MOSAIC model (phospholipid bilayer + proteins)



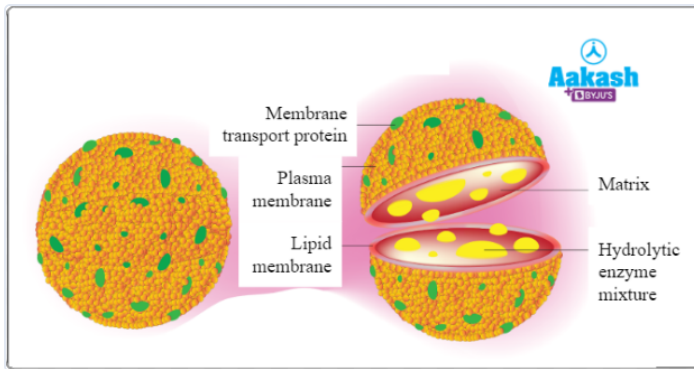
ER: rough (ribosomes = protein) vs smooth (lipid + detox)



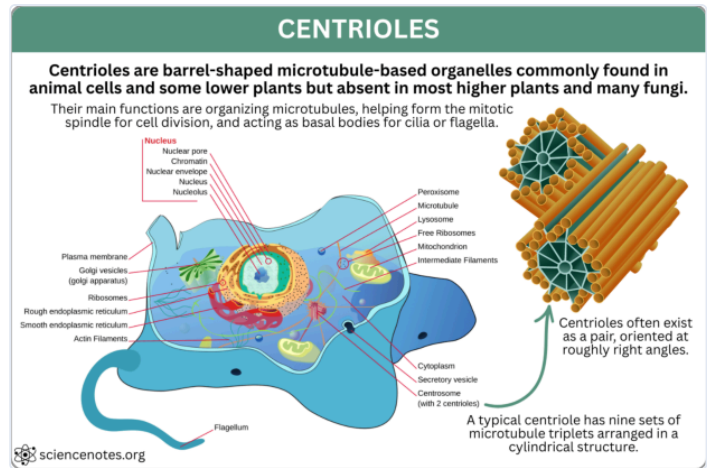
Golgi apparatus (dictyosome): cis→trans, packages & ships



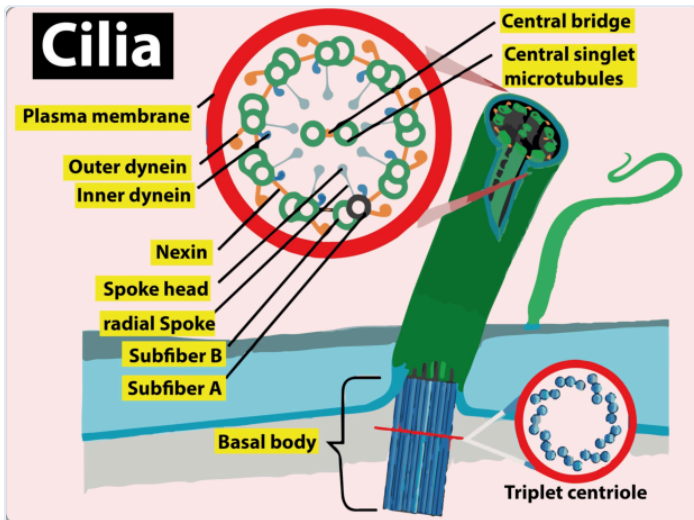
Lysosome: ~40 acid hydrolases, pH 5; hetero- + autophagy



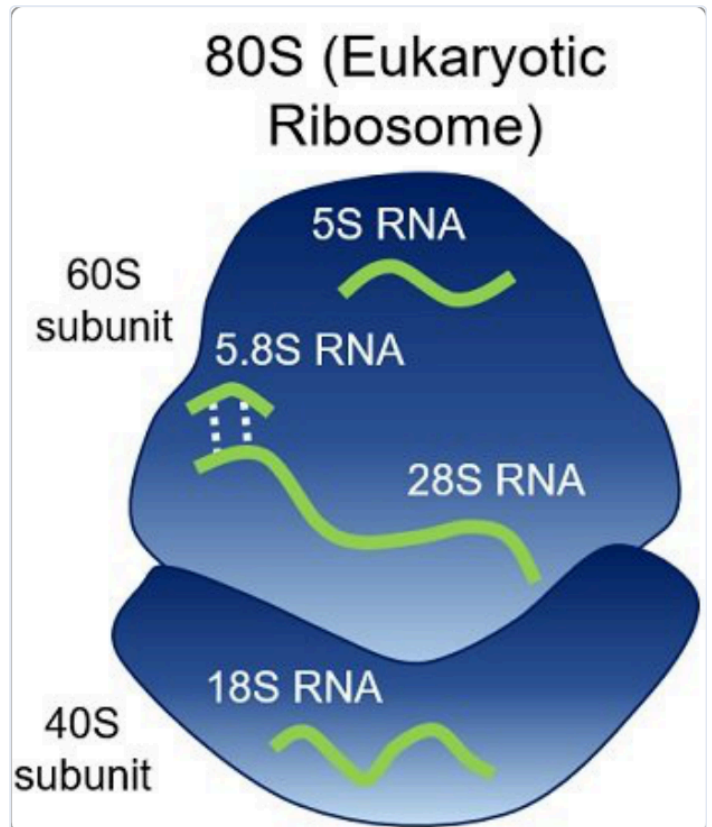
Peroxisome / microbody (catalase kills H_2O_2 ; plant = glyoxysome)



Centriole = 9 microtubule TRIPLETS (animal; makes spindle)



Cilium / flagellum = 9+2 AXONEME (9 outer doublets + 2 central)



70S vs 80S ribosome (30S+50S vs 40S+60S — NOT additive)

Similarity / Difference / Synonym: Mitochondrion & chloroplast BOTH have 2 membranes + own DNA + 70S = "semi-autonomous". DIFF: mito = respiration (both cells); chloro = photosynthesis (plant only). SYNONYMS: chondriosome = mitochondrion · dictyosome = Golgi (plant) · ergastosome = RER · tonoplast = vacuole membrane.

PROBABLE 4-MARK — "State 4 differences between a plant & animal cell"

[4]

- Plant has a CELLULOSE cell wall; animal has no wall (only membrane).
- Plant has CHLOROPLASTS (photosynthesis); animal does not.
- Plant has ONE large central VACUOLE; animal has small / many vacuoles.
- Animal has CENTRIOLES; plants generally do not. → 1 mark each = 4.

4.8-MARK EXTENSION + Add: "Both have a nucleus, mitochondria, ER, Golgi and ribosomes (common); the plant cell wall and chloroplast give it a rigid, photosynthetic form, whereas the animal cell relies on the cytoskeleton for shape." + draw & label the plant cell figure (+0.8 for the labelled diagram).

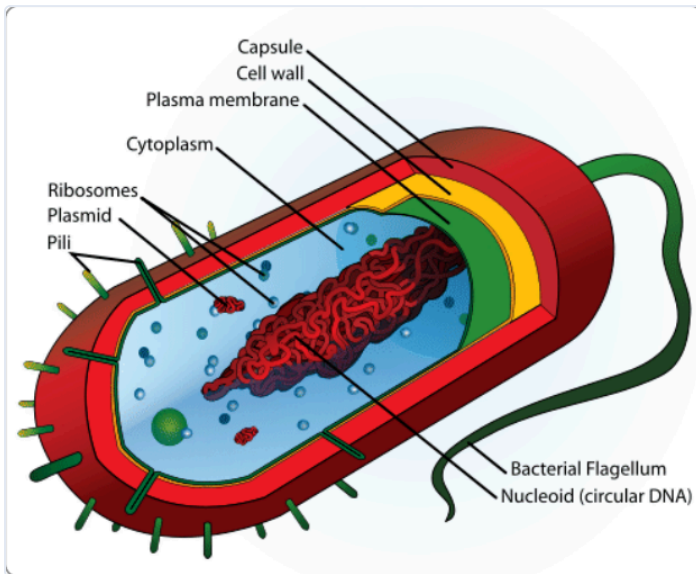
PROBABLE 4-MARK — "Draw & label the fluid-mosaic model; state 2 functions"

[4]

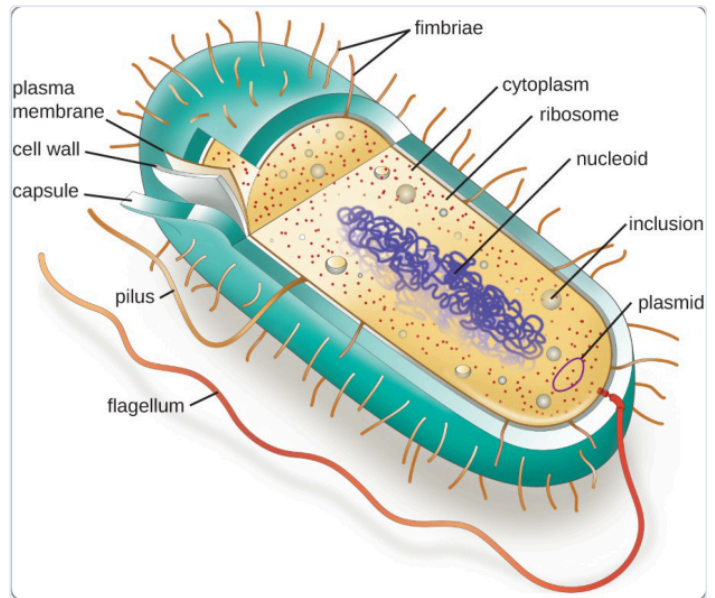
- Phospholipid BILAYER = amphipathic heads out, tails in (drawing = 1).
- Proteins: integral (spanning) + peripheral; cholesterol present (label = 1).
- Function 1: SELECTIVE PERMEABILITY (controls what enters/leaves) = 1.
- Function 2: receptor + transport + cell-recognition sites = 1.

2 — BACTERIA — figure + probable answers

What this means (plain): Bacteria = the smallest living things (prokaryotes). They have no nucleus — just a loop of DNA (nucleoid) floating in the cell — and their cell wall is made of murein (not cellulose). They multiply by simply pinching in two (binary fission).



Bacterial characteristics + shapes (coccus, bacillus, vibrio, spiral)



Unique prokaryote features: murein wall, 70S ribosomes, plasmids

Similarity / Difference / Synonym: Bacteria vs Archaea (both prokaryote): bact = murein + ester lipids; archaea = pseudomurein + ether lipids. SYNONYMS: nucleoid = genophore = incipient nucleus · murein = mucopeptide = peptidoglycan · mesosome = bacterial "respiration fold".

PROBABLE 4-MARK — "List 4 structural features of bacteria that mark them as PROKARYOTES"

[4]

- No true nucleus — a naked circular DNA (nucleoid/genophore) = 1.
- No membrane-bound organelles (no mito, no Golgi, no ER) = 1.
- 70S ribosomes (30S + 50S, non-additive) = 1.
- Rigid wall of MUREIN/peptidoglycan (NAG + NAM; absent in Mycoplasma) = 1.

4.8-MARK EXTENSION + "Cytological note": mesosomes = infoldings of the plasma membrane that substitute for mitochondria in respiration; gas vacuoles give buoyancy in water; plasmids carry antibiotic-resistance genes. Draw the bacterial cell and label nucleoid, wall, mesosome, flagellum, plasmid (+0.8).

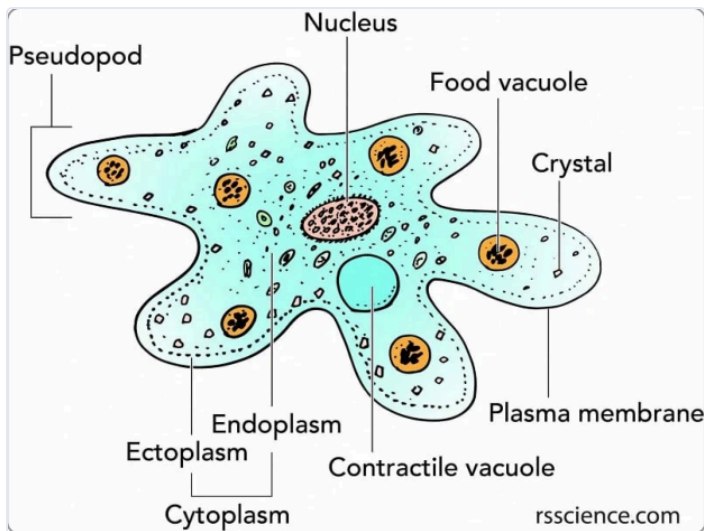
PROBABLE 4-MARK — "Distinguish Gram + and Gram - (4 points)"

[4]

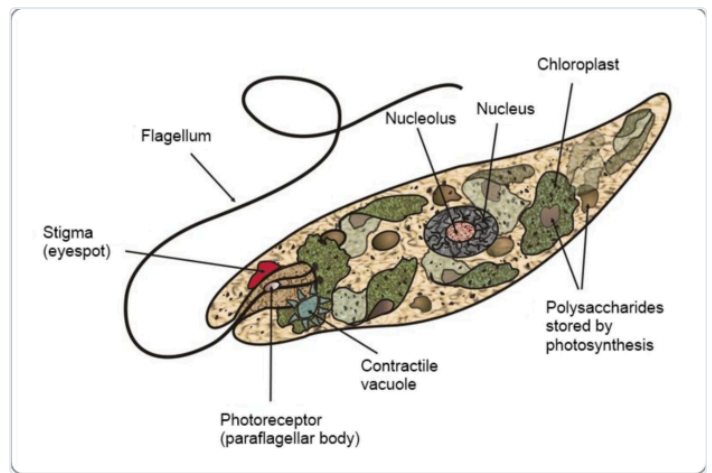
- Gram +: THICK murein (20–80 nm) + teichoic acid; retains crystal violet = BLUE.
- Gram -: THIN murein (5–10 nm) + outer LPS membrane; takes safranin = PINK.
- Gram - LPS = endotoxin (harder for antibiotics to penetrate).
- Example: Gram + = *Bacillus/Streptococcus*; Gram - = *E. coli/Vibrio*. (1 mark each = 4)

3 — KINGDOM PROTISTA (6 classes) — figures + probable answers

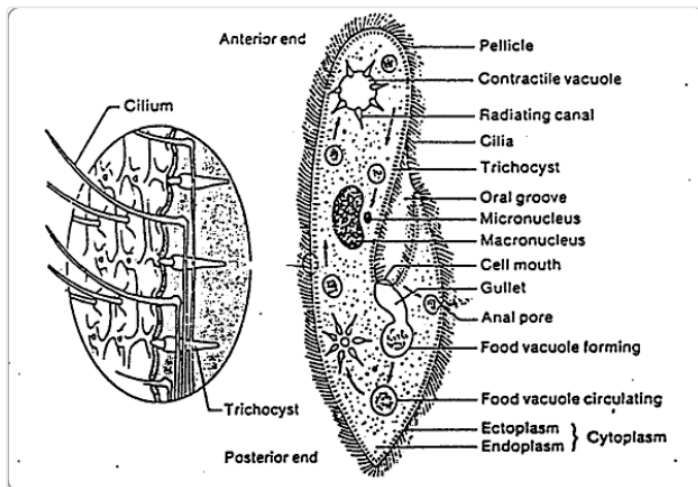
What this means (plain): Protista = the "leftover" eukaryotes that are NOT plants, animals or fungi — mostly single-celled pond-water life. The exam sorts them by HOW THEY MOVE: amoeba (pseudopod), euglena/trypanosoma (flagellum), paramecium (cilia), plasmodium (no movement / spore).



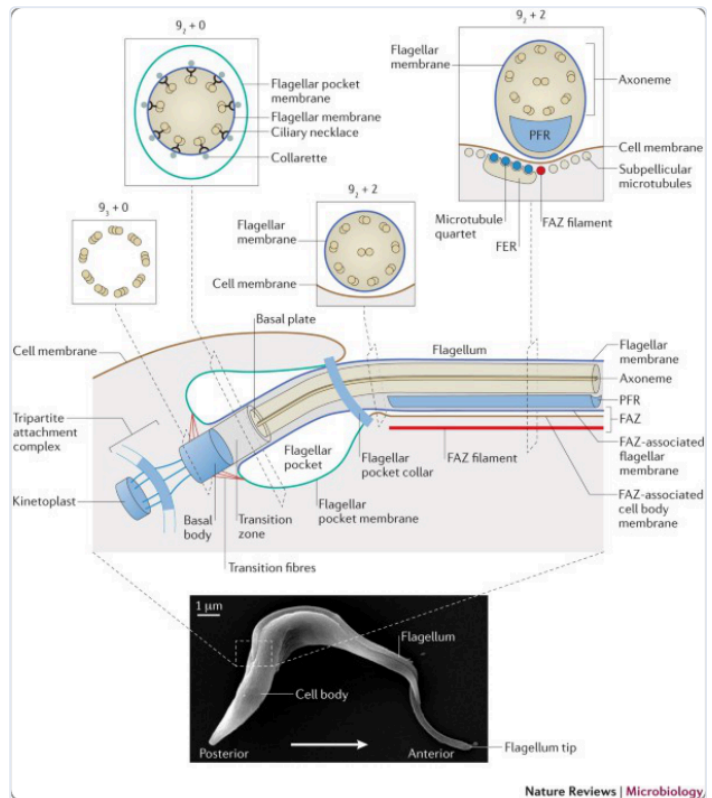
SARCODINA — *Amoeba proteus* (pseudopodia, food vacuole)



MASTIGOPHORA — *Euglena* (flagellum + eyespot + chloroplast = mixotroph)



CILIOPHORA — *Paramecium* external + internal (2 nuclei, cilia, oral groove)



Mastigophora — *Trypanosoma* (1 flagellum + undulating membrane, parasite)

Similarity / Difference / Synonym: All 4 animal-like classes are eukaryote + unicellular, they differ by locomotion only (P-F-C-N). **SYNONYMS:** Rhizopoda = Sarcodina · Flagellata = Mastigophora · Infusoria = Ciliata · Apicomplexa = Sporozoa.

PROBABLE 4-MARK — “Distinguish the 4 protozoan classes by locomotion”

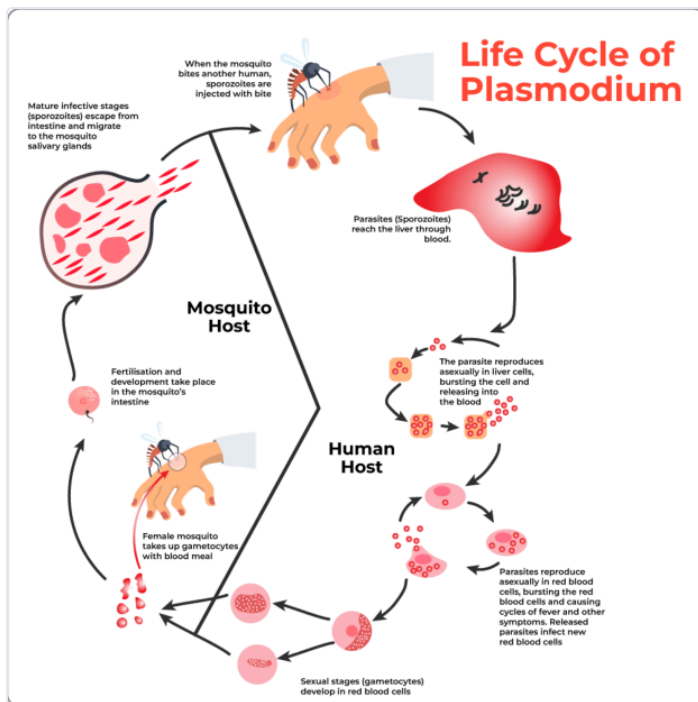
[4]

- SARCODINA: by PSEUDOPODIA (false feet; e.g. *Amoeba*) = 1.
- MASTIGOPHORA: by FLAGELLA (whip-like; e.g. *Euglena*, *Trypanosoma*) = 1.
- CILIOPHORA: by CILIA (short, many; e.g. *Paramecium*; 2 nuclei) = 1.
- SPOROZOA: NO locomotion — obligate parasite, spore (e.g. *Plasmodium*) = 1.

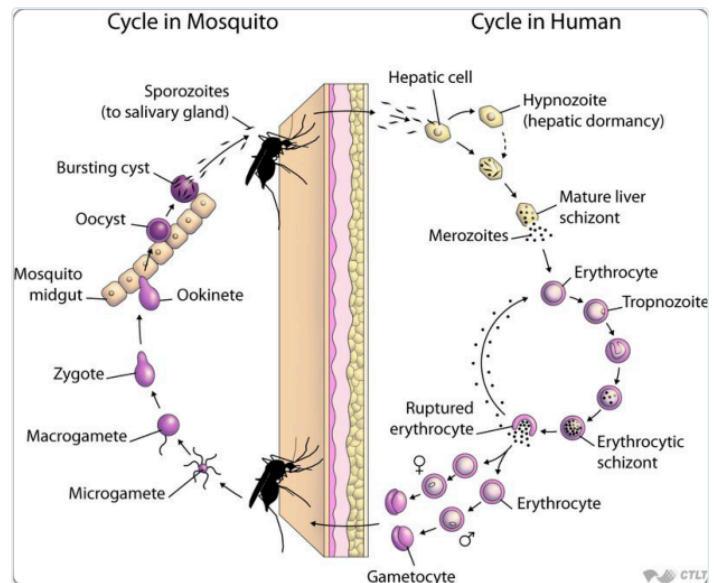
4.8-MARK EXTENSION + nutrition type of each (holozoic / osmotrophic / mixotrophic) + one disease (*Entamoeba* → dysentery; *Trypanosoma* → sleeping sickness/Chagas; *Plasmodium* → malaria) and name the 2 ALGAL divisions. Draw one labelled amoeba or euglena (+0.8).

4 — PLASMODIUM (Malaria) — figure + probable answers

What this means (plain): Plasmodium = the malaria parasite. It needs TWO hosts: you (human, where it multiplies asexually) and the female *Anopheles* mosquito (where it reproduces sexually). The mosquito's bite injects sporozoites into your blood → liver → red cells → the chills-and-fever attack.



PLASMODIUM full life cycle (human asexual + mosquito sexual) — the 7-step exam diagram



Alternate labelled malaria life cycle (cross-check the stage names)

Similarity / Difference / Synonym: Asexual = schizogony/sporogony in human; Sexual = gametogony in mosquito. **SYNONYMS:** trophozoite = "feeder" · schizont = "segmenter" · merozoite = asexual daughter · sporozoite = mosquito→human. **DIFF:** falciparum (malignant, no hypnozoite) vs vivax (benign, relapse via hypnozoite).

PROBABLE 4-MARK — "Outline 4 stages of the human (asexual) phase"

[4]

- Sporozoites enter blood on mosquito bite = 1.
- Hepatic phase: multiply in liver → merozoites (pre-erythrocytic, latent) = 1.
- Erythrocytic: merozoite → trophozoite → segmented schizont → RBC bursts = 1.
- Burst = chills + fever + anaemia; some → gametocytes (for mosquito) = 1.

4.8-MARK EXTENSION + mosquito (sexual) phase: gametocyte → zygote → ookinete → oocyst → sporozoites → salivary glands → reinfects. State "digenetic / host alternation" + "vector = Anopheles only". Draw the 7-step cycle and label human vs mosquito boxes (+0.8).

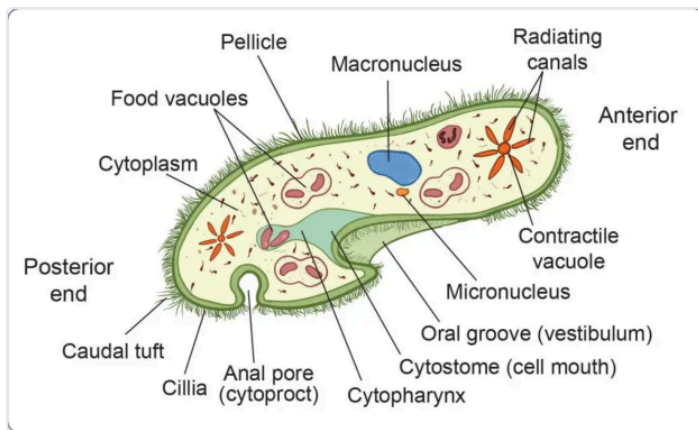
PROBABLE 4-MARK — "Differentiate 4 Plasmodium species by fever + RBC"

[4]

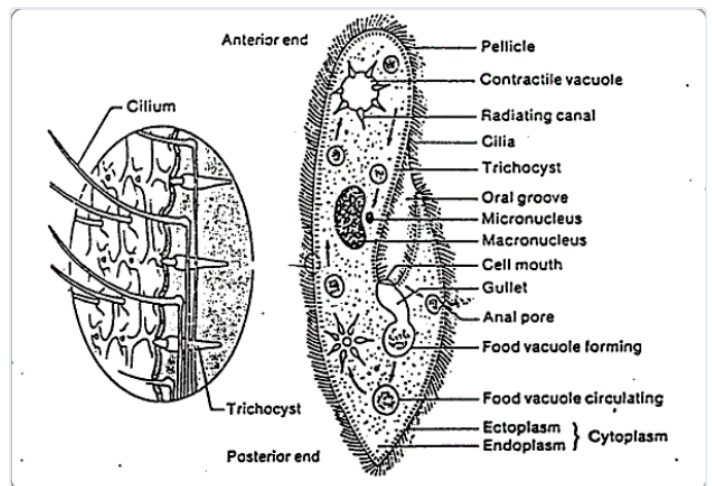
- *P. falciparum* = irregular/malignant; crescent microgametocyte; no Schuffner's dots; deadliest.
- *P. vivax* = tertian 48 h; invades OXYGENATED RBC; hypnozoite → relapse.
- *P. ovale* = tertian 48 h; invades YOUNG RBC; hypnozoite → relapse.
- *P. malariae* = quartan 72 h; rosette/daisy schizont; milder. (Mnemonic QTTK.)

5 — PARAMECIUM — figure + probable answers

What this means (plain): Paramecium = a slipper-shaped pond ciliate. Its signature is TWO nuclei (big MACRo = body/food; tiny MICRo = for conjugation/sex) + cilia (movement + sweeping food to the oral groove) + 2 contractile vacuoles (pump out extra water). It reproduces by simple fission (1→2) and by conjugation (mixing genes, NOT making more cells).



Paramecium caudatum cell structure (2 nuclei, cilia, oral groove, vacuoles)



Paramecium external + internal features (label the same 6 parts)

Similarity / Difference / Synonym: Binary fission = 1 cell → 2 clones (asexual). Conjugation = 2 cells → 2 new VARIETIES (sexual, reciprocal). SYNONYMS: macro = "body/management" nucleus · micro = "germ/meiosis" nucleus · encystation = resting cyst = hypnospore = metabola · cytopharynx = oral groove = "mouth".

PROBABLE 4-MARK — "Draw & label Paramecium; state the function of 4 labelled parts"

[4]

- Cilia = locomotion + drive food to the oral groove = 1.
- Oral groove / cytopharynx = food entry = 1.
- 2 Contractile (pulsatile) vacuoles = expel excess water (osmoregulation) = 1.
- 2 Nuclei: MACRo (somatic) + MICRo (reproductive) — its unique mark = 1.

4.8-MARK EXTENSION + food vacuole (digestion) + cytopyg (waste exit) + holozoic nutrition + osmoregulation in fresh water. Draw the cell and label all 6 (cilia, oral groove, 2 nuclei, 2 CV, food vacuole, cytopyg) (+0.8).

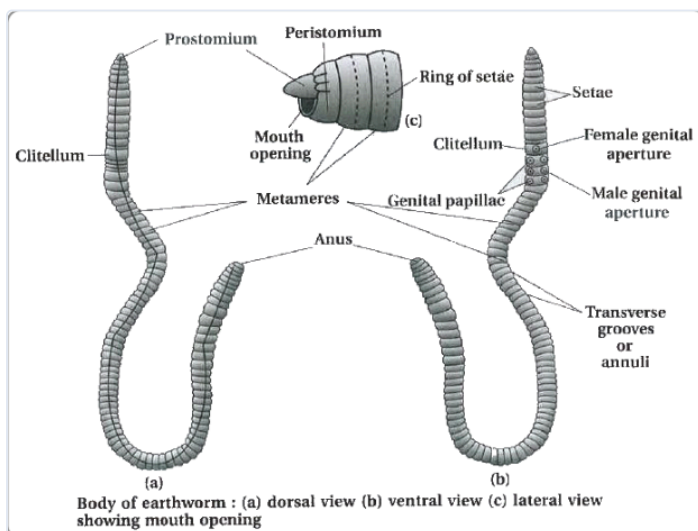
PROBABLE 4-MARK — "Describe 3 modes in the Paramecium life cycle & their role"

[4]

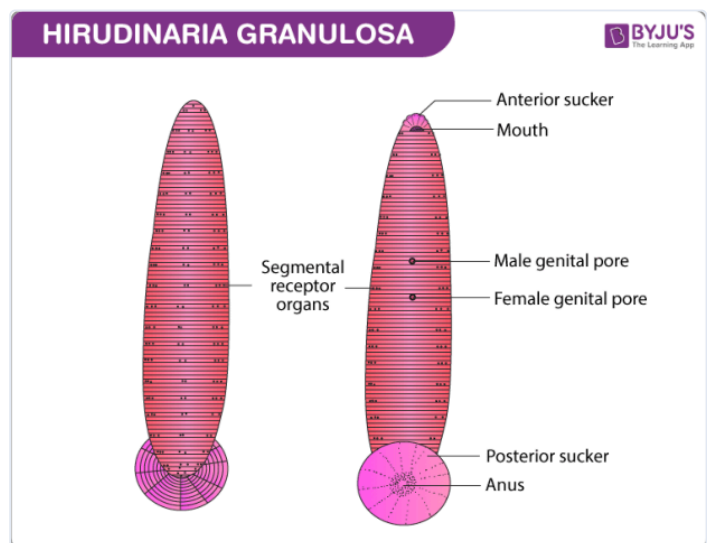
- Asexual BINARYFISSION → 2 identical daughters = rapid population = 1.
- SEXUAL CONJUGATION → reciprocal haploid exchange → new MACRo (genetic mix; not more cells) = 1.
- ENCYSTATION (hypnospore/metabola) → resting survival in dry water = 1.
- Germination of the cyst on re-watering → active ciliate = 1. (Mnemonic F-C-E.)

6 — ANNELIDA (Segmented worms) — figures + probable answers

What this means (plain): Annelida = the SEGMENTED worms (earthworms, leeches, polychaetes). Their key idea is metamerism — the body is a stack of repeated rings (segments), each with its own organs. They have a true coelom (a fluid-filled body cavity) that acts like a hydrostatic skeleton (pressure lets them burrow). Earthworms + leeches are hermaphrodites.



Earthworm (*Pheretima*) EXTERNAL structure — segments, clitellum, setae



LEECH (*Hirudinea*) — 21 segments, oral + caudal suckers, no parapodia

Similarity / Difference / Synonym: All annelids = metamerism + true coelom + triploblastic + protostome. DIFF by class: Polychaeta = parapodia + many setae + separate sexes + trochophore larva; Oligochaeta = few setae, no parapodia, hermaphrodite, no larva; Hirudinea = suckers, 21 segments. SYNONYMS: metamer = segment · coelom = true body cavity (schizocoel) · clitellum = "saddle" (cocoon gland) · aortic arches = earthworm "hearts".

PROBABLE 4-MARK — “List 4 characteristics that define Phylum Annelida”

[4]

- Metameric (SEGMENTED) body = 1.
- TRUE coelom (schizocoel) acting as hydrostatic skeleton = 1.
- Three germ layers (triploblastic) + bilateral + protostome = 1.
- Closed circulatory system + nephridia per segment = 1.

4.8-MARK EXTENSION + 3 classes with one example (*Polychaeta-Nereis*; *Oligochaeta-Pheretima*; *Hirudinea-Hirudo*); respiration by body surface (terrestrial) vs parapodia-gills (marine). Draw & label the earthworm (segments, clitellum 34-37, setae) (+0.8).

PROBABLE 4-MARK — “Earthworm (*Pheretima*): name 4 identifying anatomical facts”

[4]

- CLITELLUM (saddle / copulatory gland) on segments 34-37 = 1.
- ~100-150 segments; prostomium (lobe) before segment 1 = 1.
- SETAE (bristle bundles, 4/segment) for burrowing = 1.
- Aortic arches (“hearts”) on segments 7-11; hermaphrodite (cross-fertilisation only) = 1.

PROBABLE 4-MARK — “Distinguish Polychaeta, Oligochaeta & Hirudinea (4 points)”

[4]

- Parapodia: present (poly) / absent (oligo + hiru) = 1.
- Setae: many (poly) / few (oligo) / few-none (hiru, uses suckers) = 1.
- Sexes: separate + trochophore larva (poly) / hermaphrodite, no larva (oligo + hiru) = 1.
- Examples: *Nereis* / *Pheretima* / *Hirudo* — habitat marine / soil / water = 1.

APPENDIX D — RAPID-FIRE ONE-PAGE DUMPS (6 pages)

Each topic below is compressed into a SINGLE full-page box of “outlets” — every key fact/branch as a short line. Use these for the night-before rapid recall (read one page per topic).

HOW TO USE
Sec.1 → Sec.6 in order, one page each: read the CORE line, then every outlet line aloud, then close the page and re-scribble what you remember. Anything you miss → mark for one more pass.

Sec.1 CELL BIOLOGY — Every Single Outlet (One-Page Dump)

CORE: CELL = basic structural + functional unit of life (Cell Theory) split: PROKARYOTE (no nucleus) vs EUKARYOTE (true nucleus) — Hooke 1665 · Schleiden 1838 · Schwann 1839 · Virchow "Omnis cellula e cellula"

OUTLET A — PROKARYOTE (Bacteria/Archaea)

- Nucleoid (genophore): circular DNA, NO histone, no membrane
- 70S ribosome = 30S + 50S (S-values NOT additive)
- Wall = murein/peptidoglycan (NAG + NAM); mycoplasma: none
- No mito/Golgi/ER/plastid; respiration via MESOSOME folds
- Flagellum T.S. = 9 peripheral granules (no 9+2)
- Plasmids (extra circular DNA); gas vacuoles (aquatic)
- Divide by BINARY FISSION only (no mitosis/meiosis)
- Size 0.2–10 µm; Mycoplasma = smallest (0.1–0.3 µm)
- Archaea: ether lipids, pseudomurein, no murein — no cytokinesis

OUTLET B — EUKARYOTE (Plant/Animal/Fungi/Protist)

- True nucleus: double envelope + pores (300–1000) + nucleolus
- 80S cytoplasmic ribosome (40S+60S); 70S in mito + chloroplast
- Full organelle suite: mito, RER/SER, Golgi, lysosome, vacuole
- Plant: wall (cellulose) + plastids + big central vacuole + glyoxysome
- Animal: no wall, no plastid; CENTRIOLE (9-triplet) + abundant lysosome
- Fungus: wall = chitin — flagellum T.S. = 9+2 microtubules
- Size 10–100 µm; ostrich egg = largest (8–12 in); nerve = longest
- Division = MITOSIS (somatic) / MEIOSIS (germ)
- Exception: mature mammal RBC enucleate (camel RBC = nucleated)

OUTLET C — MEMBRANE PERMEABILITY LADDER

- Permeable: solvent + solute both (cell wall)
- Impermeable: nothing (polythene)
- Semipermeable: solvent only (parchment / filter paper)
- Selective / differentially permeable: solvent + chosen solutes 1–15 µm
- = PLASMA MEMBRANE of every living cell (Loewy & Siekevitz 1963)
- Passive = diffusion / OSMOSIS; active = ATP pumps against gradient
- Cell coagulates at 60 °C (protoplasm limit); pH slightly acidic
- Huxley: protoplasm = "physical basis of life" · Strasburger: cytoplasm
- Ergastic inclusions = reserve food / secretory / excretory (non-living)

OUTLET D — CELL CYCLE G1 → S → G2 → M

- G1: growth + protein synthesis; exit to G0 (permanent stop)
- S: DNA replicated; chromosomes 46, CHROMATIDS 92, 4C DNA
- G2: final growth + spindle prep; centrosomes migrate
- M: PMAT + cytokinesis (plant = cell-plate; animal = furrow)
- Daughter = 2 identical 2n cells (2C each)
- Checkpoints: G1 (size/DNA) · G2 (ready?) · Metaphase SAC
- Checkpoint failure → apoptosis / cancer (uncontrolled division)
- G0 examples: neurons, cardiomyocytes, mature RBC (can't divide)
- Amitosis = direct division (liver, some protists; unreliable)

OUTLET E — ORGANELLE ONE-LINERS (N-M-G-L-R-C-V)

- Mitochondria = ATP; cristae + oxysome (F1, Lehninger) + 70S + DNA
- Chloroplast = light rxn in GRANUM + dark rxn in STROMA; quantasome 230–250 Chl
- RER = protein (ribosomes); SER = lipid + detox (no ribosomes)
- Golgi = modify/package/ship; plant name = DICTYOSOME
- Lysosome = ~40 hydrolases, pH 5; 4-stage life (primary → hetero → residual → autophagosome)
- Peroxisome = catalase kills H₂O₂; plant = GLYOXYISOME (glyoxylate cycle)
- Ribosome = 23 nm protein engine; POLYSOMES ride on mRNA
- Vacuole = TONOPLAST membrane = same as plasma membrane; turgor
- Centriole = 9-triplet; plants: absent; animals: mandatory
- Semi-autonomous: mito + chloroplast have own DNA + 70S but need NUCLEUS

OUTLET F — SCALES + CELL-THEORY DEMERITS

- Naked-eye minimum = 100 µm — most cells 10–100 µm
- Light microscope resolution floor = 0.3 µm (2D image)
- TEM = 2–10 nm (2D); SEM = 5–20 nm (3D surface)
- RBC ~ 25 trillion — WBC ~ 30 billion
- Demerit 1: sclerenchyma / xylem = dead, no protoplasm but function
- Demerit 2: *Rhizopus* / *Vaucheria* = coenocytic (no wall, multinucleate)
- Demerit 3: virus = alive-ish (replicates) but no cell, no protoplasm
- Demerit 4: mature RBC + phloem sieve tube = alive but enucleate
- 10 tenets of modern cell theory (must name 3: composition, arise, ...)
- Cytocinesis = rotation (1 vacuole) vs circulation (many vacuoles)

MNEMONICS + EXAM GUTS

- Organelle one-liners "N-M-G-L-R-C-V-C-P" = Never Make Green Lizards Run Cold Vibrating Caged Pigeons
- Cell theory "H-S-V" = Hooke (term) · Schleiden+Schwann (plant+animal) · Virchow (from cells)
- Microscope pioneers "R-L-J-K-Z" = Hooke (first) · Leeuwenhoek (simple) · Janssen (compound) · Knoll & Ruska 1931 (electron) · Zernike (phase contrast)
- Protoplasm theories "P-F-H-A-B-W" = Purkinje + Von Mohl (term) · Fleming (fibrillar) · Hanstein (protoplast) · Altmann (granular) · Bütschli (foam) · Wilson
- Svedberg non-additive: 30+50 = 70 (not 80) · 40+60 = 80 (not 100)
- C-value track: G1 2C → S 4C → M daughter 2C each · meiosis final gamete = C (half somatic)
- GOTCHA: "cell = protoplasm enclosed by selective permeable membrane" = Loewy & Siekevitz 1963 (exam quote)
- GOTCHA: Camel RBC keeps nucleus; mammal RBC doesn't; sieve-tube cell alive but enucleate

Figure 1: Cell Biology — all single outlets on one page (6 outlet boxes + mnemonic strip)

Sec.2 BACTERIA — Every Single Outlet (One-Page Dump)

CORE: Bacteria = PROKARYOTIC, ubiquitous, smallest life (~0.2–10 µm) — Kingdom Monera = Bacteria + Cyanobacteria + Archaea — “decomposers and mineralisers of the biosphere”

OUTLET A — IDENTITY CARD (5 points)

- Prokaryote: no true nucleus, no membrane-bound organelle
- Nucleoid = circular DNA + RNA + protein; no histones
- 70S ribosomes (30S+50S) — no Golgi / mito / plastid / ER
- Wall = murein = N-acetyl-glucosamine + N-acetyl-muramic acid (NOT cellulose)
- Nutrition auto/hetero; gas vacuole for buoyancy; no true vacuole
- Reserve food = glycogen / fats; flagellum = flagellin protein
- Reproduction = binary fission; sexual = absent; recombination = CTC only
- Examples: *Vibrio*, *Diplococcus*, *Bacillus*, *Streptococcus*, *E. coli*
- *Mycoplasma* = no wall + pleomorphic + penicillin-resistant + smallest

OUTLET B — SHAPE CLASSIFICATION (Cohn)

- Cocci (round): diplococcus (2) · streptococcus (chain) · staphylococcus (cluster)
- Bacilli (rod): diplobacillus · streptobacillus · single · with/without flagella
- Vibrio (comma): *V. cholerae* (cholera)
- Spirillum (rigid spiral): *S. volutans* = longest prokaryote (15 µm)
- Spirochaete (flexible spiral): *T. pallidum* = syphilis
- Cyanobacteria (“blue-green”): *Nostoc*, *Anabaena* (heterocyst fixes N₂)
- *Mycoplasma* = irregular / pleomorphic (no wall so shape-free)
- Arrangement memory: D-S-T = Diplo (2) · Strepto (chain) · Tetrads (4) / sarcinae (8)
- “Joker of the plant kingdom” = *Mycoplasma* (smallest + no wall)

OUTLET C — GRAM STAIN + WALL TYPES (4)

- Gram + = thick murein 20–80 nm + teichoic acid → takes crystal violet = BLUE
- Gram – = thin murein 5–10 nm + outer LPS membrane → takes safranin = PINK
- Acid-fast = *Mycobacteria* (waxy mycolic acid); Ziehl-Neelsen = red; TB / leprosy
- No-wall = *Mycoplasma* (pleomorphic + osmotic lysis risk)
- LPS = endotoxin of Gram – (septic shock risk)
- Penicillin attacks murein synthesis → kills Gram+ readily; Gram– harder
- Capsule = anti-phagocytic layer (*Klebsiella*, *S. pneumoniae*)
- Memory “G-P thick, G-N thin+outer” + “Murein = NAG + NAM (not cellulose)”
- Plasmid = antibiotic-resistance + gene transfer vehicle (F-plasmid = fertility)

OUTLET D — REPRODUCTION + RECOMBINATION (5)

- Binary fission (primary asexual): DNA copies → septum → 2 clones
- Endospore (survival, NOT reproduction): *Bacillus*, *Clostridium*; heat-resistant
- Conjugation: F⁺ donates F-plasmid via sex pilus → F[–] becomes F⁺
- Transformation: uptake of naked DNA from environment (Griffith's experiment)
- Transduction: bacteriophage carries donor DNA into new host
- Mnemonic “CTC” = Conjugation · Transformation · Transduction
- No syngamy, no meiosis, no true sex in bacteria
- Generation time = ~20 min for *E. coli* (exponential 1→2→4→8)
- Mesosome = respiration site (mitochondria-equivalent)

OUTLET E — BACTERIA vs ARCHAEA (6 differences)

- Wall: murein (bact) vs pseudomurein (arch)
- Membrane lipids: ester bonds (bact) vs ether bonds (arch)
- Methanogen? Arch only (methane producer)
- Extremophile? Common in arch (thermophiles, halo, acidophiles)
- Antibiotic sensitivity: bact = yes; arch = generally no
- RNA polymerase: bact = 1 type; arch = eukaryote-like (multi-subunit)
- Habitat: bact = everywhere; arch = mostly extreme (hot springs, salt lakes)
- Exam gotcha: “all bacteria are prokaryotic” = TRUE; “all protists are eukaryotic” = TRUE
- “Cyanobacteria = blue-green algae” = TRUE (but they are bacteria, not algae)

OUTLET F — BACTERIAL PATHOGENS + APPLICATIONS (exam basis)

- Pathogens: *M. tuberculosis* (TB) · *V. cholerae* (cholera) · *S. pneumoniae*
- Beneficial: *Lactobacillus* (curd) · *Rhizobium* (N₂ fixation in root nodules)
- Industrial: *S. cerevisiae* (yeast/yogurt) · *Acetobacter* (vinegar) · *Streptomyces* (antibiotics)
- Nitrogen cycle: *Azotobacter* + *Rhizobium* + cyanobacteria = N₂ fixers
- Antibiotics target: cell wall (penicillin), ribosome (tetracycline), DNA
- Endospore = NOT reproduction: 1 cell → 1 spore → 1 cell when water returns
- “Joker of plant kingdom” = *Mycoplasma* (smallest, no wall)
- “Bacteria = smallest living thing” = FALSE: viruses are smaller (but not fully alive)
- “*Mycoplasma* = smallest LIVING” = TRUE
- Plasmid + transposon = “jumping genes” + resistance transfer (e.g. R-plasmid)

MNEMONICS + EXAM GUTS + ONE-PAGE RECAP

- Shape “C-D-S-T” = Coccus → Diplo- → Strepto- → Staphylo- → Tetrads
- Recombination “CTC” = Conjugation (pilus) · Transformation (naked DNA) · Transduction (phage transit)
- Wall “G-P thick, G-N thin + LPS outer, acid-fast waxy, no-wall mycoplasma”
- “Murein = NAG + NAM (not cellulose)” + “70S = 30S + 50S (not additive)”
- Bacteria vs archaea: “ester / ether / methanogen / extremophile” = archaea signature
- Endospore ≠ reproduction (1→1→1); binary fission = asexual (1→2)
- EXAM GUT: “no true sex” in bacteria = asexual binary fission + 3 CTC recombination events
- EXAM GUT: “murein” = defining marker of bacterial cell wall (absent in mycoplasma, archaea, eukaryotes)

Figure 2: Bacteria — all single outlets on one page

Sec.3 KINGDOM PROTISTA — Every Single Outlet (One-Page Dump)

CORE: Protista = eukaryotic “residuals” (not plant/animal/fungus); unicellular, mostly microscopic, free-living or parasitic — Haeckel 1869 coined Protista — 4 animal-like classes + algal divisions (check YOUR book’s exact 6)

CLASS 1 — SARCODINA (a.k.a. Rhizopoda / Amoebida)

- Locomotion = pseudopodia (“false feet”, cytoplasmic streaming)
- Irregular shape; 1 or many nuclei; NO rigid covering
- Nutrition = phagocytosis (engulf → food vacuole + lysosome)
- Forms cysts (resting spores) in dry / harsh conditions
- Ex.: *Amoeba proteus*, *Entamoeba histolytica*, *Arcella*, *Polytoma*
- Identify: “irregular crawling + pseudopod” = Sarcodina
- Parasitic form: *Entamoeba* = amoebiasis/dysentery (liver abscess)
- Reproduction: binary fission + conjugation (some)

CLASS 2 — MASTIGOPHORA (a.k.a. Flagellata / Zoomastigina)

- Locomotion = FLAGELLA (whip-like; 1 to many)
- Flexible PELLICLE (no rigid wall); some have chloroplasts
- Mixotrophic = photosynthesis + heterotrophy (*Euglena*)
- Red “eyespot” (stigma) = light sensing
- Ex.: *Euglena*, *Trypanosoma*, *Giardia*, *Leishmania*, *Trichomonas*
- *Trypanosoma* = African sleeping sickness / Chagas; flagellum 1 + undulating membrane
- *Giardia* = diarrhoea; “flagellate with 2 nuclei” (unique!)
- Dinoflagellates: 2 flagella (one transverse = spin); bioluminescence

CLASS 3 — CILIOPHORA (a.k.a. Ciliata / Infusoria)

- Locomotion = CILIA (short, many, rhythmic, body-covering)
- TWO nuclei (di-primary): MACRO (somatic) + MICRO (germ)
- Most complex protozoan; food swept to oral groove / cytopharynx
- 2 contractile (pulsatile) vacuoles = osmoregulation
- Ex.: *Paramecium caudatum*, *Opalina*, *Stentor*, *Vorticella*, *Balantidium*
- Identify: “many cilia + 2 nuclei” = Ciliophora (see Sec.5 for full *Paramecium*)
- Reproduction: binary fission + CONJUGATION (reciprocal haploid exchange)
- Encystation = resting survival; cilia retract + coat forms

CLASS 4 — SPOROZOA (a.k.a. Apicomplexa / Coccidia)

- Locomotion = NONE in adult (obligate PARASITE; spore dispersal)
- Apical complex (rhoptries, polar rings) = host-cell invasion
- Digenetic = 2 hosts; sexual + asexual alternation
- Spore-forming (sporangium); reduced organelles
- Ex.: *Plasmodium* (malaria), *Toxoplasma gondii*, *Cryptosporidium*, *Isospora*
- Identify: “parasite + spore + no cilia/flagella” = Sporozoa (see Sec.4)
- *Toxoplasma* = cat-spread; dangerous in pregnancy (miscarriage risk)
- Asexual = schizogony / sporogony; sexual = gametogony

DIVISION 5 — ALGAL: CHRYSOPHYTA / HAPTOPHYTA (golden-brown)

- Golden-brown (fucoxanthin + Chl a,c); photosynthetic
- Diatoms have a siliceous TEST (glass shell; SiO₂)
- Some have calcium-carbonate statoliths (balance)
- Ex.: *Prymnesium*, *Diatoma*, Chrysophyceae
- Identify: “golden-brown pigment + siliceous shell” = Chrysophyta
- Diatomaceous earth = fossil diatom shells (filter / abrasive)
- Major marine primary producers (oxygen + carbon sink)
- Cell wall = cellulose OR siliceous test (not peptidoglycan)

DIVISION 6 — CYANOPHYTA / CYANOBACTERIA (blue-green)

- Photoautotrophic PROKARYOTES (a “bacteria” in the protist list)
- 70S, no true nucleus, murein wall, photosynthetic pigments
- HETEROCYSTs fix N₂ (*Anabaena*, *Nostoc*)
- Store food = cyanophycean starch / cyanophycean vacuole
- Ex.: *Nostoc*, *Anabaena*, *Oscillatoria*, *Spirulina*
- EXAM TRAP: cyanobacteria are BACTERIA, NOT true algae / eukaryotes
- Blooms = “water turning red/brown”; often toxic to fish
- Some euglenoid lists use Euglenophyta instead of cyanophyta — check your book

LOCOMOTION 4-WAY (P-F-C-N) & NUTRITION 4-TYPES

- P = Pseudopodia (Sarcodina, slow, direction-shifting)
- F = Flagella (Mastigophora, fast, whip-like)
- C = Cilia (Ciliophora, rhythmic, reversible, many)
- N = None (Sporozoa; drift / host carries) · mnemonic “P-F-C-N”
- Holozoic = ingest (*Paramecium*, *Amoeba*) · Osmotrophic = absorb (slime mould, fungi-like)
- Photosynthetic = make (algae) · Mixotrophic = both (*Euglena*)

ONE-LINE IDENTITY + REPRODUCTION

- Each class = 1 locomotion type + 1 nutrition type = ID key
- Asexual = binary / multiple fission; cyst = survival
- Sexual = conjugation / syngamy (genetic recombination)
- Ciliophora = only class with TWO nuclei (macro+micro)
- Sporozoa = only class that is OBLIGATE parasite
- Protista is PARAPHYLETIC (“residual” kingdom; not all one clade)
- “Proto-” = first; “zoon” = animal; “phyta” = plant-like (algal)
- Free = pond / soil / marine; parasitic = body tissues / blood
- Mnemonic “S-M-C-S” = Sarcodina-Mastigophora-Ciliophora-Sporozoa

Figure 3: Protista — all single outlets on one page (4 protozoan classes + 2 algal divisions + locomotion + nutrition)

Sec.4 PLASMODIUM — Every Single Outlet (One-Page Dump)

CORE: Plasmodium = Sporozoa / Apicomplexa obligate parasite; causes MALARIA; 2-host digenetic (human asexual + mosquito sexual)
Domain Eukaryota → Alveolata → Apicomplexa → Haemosporida → Plasmodiidae → Plasmodium (NCBI lineage)

OUTLET A — CLASSIFICATION (full ladder)

- Kingdom Protista · Phylum Protozoa (Apicomplexa) · Class Sporozoa
- Subclass Apicomplexa (Cnidospora) · Order Eucoccidiorida (Haematozoa)
- Family Plasmodiidae · Genus Plasmodium
- 5 human spp: falciparum, vivax, ovale, malariae, knowlesi
- Vector = FEMALE Anopheles ONLY (not Aedes); man-biting, nocturnal
- Hosts: human (blood/sexual cycle data) + mosquito; alternation of generations
- No cilia/flagella in adult; apical complex = invasion structure
- Obligate intracellular: liver (exoerythrocytic) + RBC (erythrocytic)

OUTLET B — 4 MORPH FORMS (identify each)

- SPOROZITE = infective, mosquito spit-gland → liver (crescentic)
- TROPHOZOITE = growing “feeding” form in RBC (ring-stage, early)
- SCHIZONT (segmented) = merozoite-packed, then RBC bursts
- GAMETOCYTE = sexual: MACRO (round, female) + MICRO (crescent, male)
- falciparum microgametocyte = BANANA / crescent shape (diagnostic)
- Merozoite = asexual daughter released from schizont
- Haemozoin (black pigment) = RBC hemoglobin waste in trophozoite
- Gametocyte = the ONLY form the mosquito can ingest (infective to mosquito)

OUTLET C — HUMAN PHASE (asexual) 5 STEPS

1. Bite: sporozoites in mosquito saliva → blood
2. Hepatic: sporozoite → hepatocyte → 1st schizogony → merozoites
3. Blood: merozoite → RBC → trophozoite → schizont → burst
- Rupture = CHILLS + FEVER + HEMOLYSIS (anaemia) = the malaria attack
4. Repeat schizogony = cycles of fever (tertian/quartan/quotidian)
5. Some trophozoites → gametocytes (ready for mosquito)
- Pre-erythrocytic = asymptomatic; erythrocytic = symptomatic
- Latency: falciparum ~9 d; vivax/ovale can be dormant (hypnozoite)
- Treatment targets trophozoite; radical cures hypnozoite relapse
- Severe = cerebral malaria (falciparum; RBCs clog brain capillaries)

OUTLET D — MOSQUITO PHASE (sexual) 4 STEPS

1. Mosquito swallows gametocytes from infected human blood
2. Microgametocyte EXFLAGELLATES → many microgametes; fertilisation
3. Zygote → motile ookinete → gut wall → OOCYST (sporogony)
4. Oocyst bursts → SPOROZITES migrate to salivary glands
- Mosquito becomes INFECTIONIVE; next bite re-inoculates sporozoites
- Sporozoites = the ONLY form passed to a new human
- Sexual = in mosquito; asexual = in human (alternation of generations)
- Control = drain stagnant water + bed-nets (stop mosquito breeding)
- Malaria = the “agony” of the periodic fever; “tertian” = every 48 h
- “Malignant tertian” = falciparum; “benign tertian” = vivax

OUTLET E — 5 SPECIES + FEVER PATTERN (QTTK)

- falciparum = irregular; most dangerous; no Schuffner’s dots
- vivax = tertian 48 h; invades OXYGENATED RBC + hypnozoite (relapse)
- ovale = tertian 48 h; invades YOUNG RBC + hypnozoite + pointed RBC
- malariae = quartan 72 h; rosette/daisy schizont; milder
- knowlesi = quotidian ~24 h; zoonotic (macaque → man)
- Mnemonic “QTTK”: Q = quartan (malariae) · T = tertian (vivax, ovale) · f = irregular · K = 24 h (knowlesi)
- Relapse = ONLY vivax + ovale (hypnozoite wakes later)
- Schuffner’s dots = faint fine stippling on RBC edge (vivax/ovale)

OUTLET F — SYMPTOMS, CONTROL, EXAM GUTS

- Classic = 3-stage paroxysm: CHILL → FEVER → SWEAT (periodic)
- Anaemia = RBC destruction; splenomegaly (enlarged spleen)
- Cerebral malaria = falciparum (delirium, coma, convulsion)
- Drug resistance (falciparum to chloroquine) = global concern
- Diagnosis = thick/thin blood film under microscope (gold standard)
- Control = DDT spraying + bed-nets + drain water + antimalarials
- EXAM GUT: “definitive host” = where the SEXUAL phase happens = mosquito (NOTE: some texts swap host names — follow YOUR book)
- EXAM GUT: “sporozoite” infects human; “gametocyte” infects mosquito

MNEMONICS + EXAM GUTS + ONE-PAGE RECAP

- Lifecycle order: “BITE→LIVER→BLOOD→GAMETO→ZW→SALIVA→BITE” (B→L→B→G→ZW→S→B)
- 4 forms = “S-T-S-G”: Sporozoite · Trophozoite · Schizont · Gametocyte (memory order in host)
- Fever “QTTK”: quartan = malariae, tertian = vivax/ovale, falciparum = irregular, knowlesi = 24 h
- “EX-FLAG-ELLATION” = microgametocyte makes many motile microgametes in mosquito gut
- Schizogony = asexual multiple-fission (human); Sporogony = zygote→spore (mosquito)
- Host alternation = human (asexual) + mosquito (sexual) = “digenetic”
- GUT: vector = female ANOPHELES only; falciparum = deadliest; vivax/ovale = relapse (hypnozoite)
- GUT: “malignant” = falciparum; “benign” = vivax; quartan = malariae; 24 h = knowlesi

Figure 4: Plasmodium — all single outlets on one page

Sec.5 PARAMECIUM — Every Single Outlet (One-Page Dump)

CORE: *Paramecium caudatum* = Ciliophora; free-living freshwater ciliate; 50–300 µm slipper-shaped
Domain Eukaryota → Alveolata → Ciliophora → Oligohymenophorea → Peniculida → Parameciidae → *Paramecium caudatum*

OUTLET A — CLASSIFICATION (full ladder)

- Kingdom Protista · Phylum Ciliophora · Subphylum Intramacronucleata
- Class Oligohymenophorea · Subclass Peniculidia · Order Peniculida
- Family Parameciidae · Genus *Paramecium*
- Species *P. caudatum* (Ehrenberg 1833/34; “caudatum” = tail-like)
- Freshwater; ponds / stagnant water; “slipper” = caudatum shape
- Unique: TWO nuclei = MACRO (somatic) + MICRO (germ/reproductive)
- Pellicle = flexible outer covering (not a rigid wall); cilia all over
- Cytoplasm = ectoplasm (outer) + endoplasm (inner, streaming)

OUTLET B — ORGANELLE / STRUCTURE OUTLETS

- Cilia = many short hairs; beat rhythmically = LOCOMOTION + food current
- Oral groove (cytopharynx) = “mouth”; food → food vacuole
- Food vacuole = digestion; waste → cytopyg (anal pore / cytocyst)
- 2 CONTRACTILE vacuoles (anterior + posterior) = osmoregulation
- MACRONUCLEUS = many copies, controls body functions (amitotic)
- MICRONUCLEUS = diploid, divides by MEIOSIS, used in conjugation
- Respiration + excretion = by diffusion through PELLICLE (no organ)
- Nutrition = HOLOZOIC (ingest → intracellular digestion)

OUTLET C — LOCOMOTION + NUTRITION

- Ciliary current = food swept from groove to cytopharynx
- Forward = cilia sweep to rear; backward = sweep reversed
- Contact/aversion reaction = turns away from obstacle (kinesis)
- Food = bacteria, yeast, small particles (holozoic)
- Digestion in food vacuole (lysosome fuses); undigested → cytopyg
- Osmoregulation: fresh-water → water enters → CV expels (turgor)
- 2 CVs: one near each end; they contract rhythmically
- In hypertonic / salt water: no CV needed; in fresh water: essential
- No respiration organ: O₂ in, CO₂ out by diffusion across pellicle
- “Slipper-paramecium” = caudatum; long tail = the “caudatum” end

OUTLET D — COMPLETE LIFE CYCLE (3 processes)

- A. BINARY FISSION (asexual):** both nuclei divide; cell splits longitudinally
— = population growth; main method in healthy pond
- B. CONJUGATION (sexual, AMALGAMATION):** 2 cells meet, exchange genetic material
— Macro fragments; micro → meiosis → conjugant swaps (reciprocal)
— Forms SYNKARYON (diploid) → new macro; NOT more cells (2 NEW VARIETIES)
- C. ENCYSTATION (survival):** cilia retract + coat + METABOLA (hypnospore)
— Survives drought; on re-watering coat dissolves → GERMINATION
- D. HYPNOSPORE → RESYNTHESIS** (new ciliate after conjugation)
— Mnemonic “F-C-E”: Fission (multi) · Conjugation (mix) · Encystation (survival)
— Encystation ≠ reproduction — it is a RESTING stage (no new cells)

OUTLET E — THE TWO-NUCLEUS RULE (identity)

- MACRONUCLEUS = somatic: multi-lobed, amitotic, cell management
- MICRONUCLEUS = germ: tiny, meiotic, conjugation + genetics
- Mnemonic “MacroM / MicroM = Management vs Meiosis”
- No macro → cell dies after a few divisions (germ needed to rebuild)
- The 2-nucleus state is the CILIOPHORA exam identifier
- Excretion = contractile vacuole (water) + cytopyg (solid waste)
- Respiration / O₂ / CO₂ = diffusion through pellicle (no organ)
- “Paramecium” = ancient “before-me” / “before you” (Latin)

OUTLET F — VARIETIES + EXAM GUTS

- Species: *P. caudatum* (long), *P. bursaria* (with *Zoochlorella* symbiosis)
- *P. bursaria* = hosts green algae INSIDE (photosynthesis + locomotion)
- Conjugates: EXCONJUGANTS = the 2 new cells after exchange
- GUT: conjugation = 2 cells → 2 cells (recombined), NOT 4 daughters
- Binary fission = 1 cell → 2 identical cells (clones)
- “Ciliate” = body “covered with cilia” (defining Ciliophora trait)
- GUT: 2 contractile vacuoles + 2 nuclei + oral groove = Paramecium
- *Bursaria* = textbook example of PROTOSYMBIOSIS (algae inside ciliate)

MNEMONICS + EXAM GUTS + ONE-PAGE RECAP

- Structure ID “2-2-1”: 2 nuclei · 2 contractile vacuoles · 1 oral groove
- Two nuclei “MacroM / MicroM” = Management (body) vs Meiosis (germ)
- Life cycle “F-C-E” = Fission (asexual) · Conjugation (sexual) · Encystation (rest)
- “Encystation ≠ reproduction” = it is survival, not more cells
- Conjugation produces NEW VARIETIES (genetic mix), not 4 daughters
- Locomotion = cilia (rhythmic, reversible); nutrition = holozoic (oral groove)
- GUT: Ciliophora = only class with TWO nuclei + cilia + oral groove (exam trio)
- GUT: *P. bursaria* = endosymbiont algae (green) = the “algae inside ciliate” question

Figure 5: *Paramecium* — all single outlets on one page

Sec.6 ANNELIDA — Every Single Outlet (One-Page Dump)

CORE: Annelida = segmented worms; ~16,500 spp; true coelom; metamerism; protostome — “annelus” = small ring (segmentation) — Earthworm <i>Pheretima</i> / <i>Lumbricus</i> = Oligochaeta exam focus	
OUTLET A — PHYLUM-LEVEL FEATURES — Triploblastic + bilateral + schizocoelomate + PROTOSTOME — TRUE coelom (mesoderm split); coelomic fluid = hydrostatic skeleton — METAMERISM = repeated segments (exterior + interior organs) — Body wall = cuticle over epidermis + circular + longitudinal muscle — Respiration = body surface (terrestrial) / parapodia-gills (marine) — Excretion = nephridia (one pair/seg); metanephridial in earthworm — Circulation = CLOSED (vessels); “hearts” = aortic arches (seg 7-11) — Nerve = VENTRAL cord + supra/sub-oesophageal ganglia (“brain”)	OUTLET B — 3 CLASSES (P-O-H) COMPARISON — POLYCHAETA (Scaphia): marine, PARAPODIA + many setae + cirri, separate sexes + TROCHOPHORE larva; ex. <i>Nereis</i> (sea horse worm) — OLIGOCHAETA: soil/freshwater, few setae, no parapodia, hermaphrodite, no larva; ex. <i>Pheretima</i> (earthworm), <i>Lumbricus</i> — HIRUDINEA: leeches; few/absent setae; SUCKERS (oral + caudal), parasitic/depredatory, fixed 21 segments, hermaphrodite; ex. <i>Hirudo</i> — Mnemonic “3-H-T”: 3 classes · Hermaphrodite (oligo+hiru) · True coelomate — “P-O-H” = Polychaeta · Oligochaeta · Hirudinea
OUTLET C — EARTHWORM (<i>Pheretima</i>) BODY — ~100-150 segments; prostomium (lobed, pre-segment 1) at front — CLITELLUM = saddle (seg 34-37) = cocoon-secreting copulatory gland — SETAE = 4 bundles/segment (anchor for burrowing; bristles) — Band of genitalia (segments ~14, 15 + 27) marks the clitellar region — EXAM ID: clitellum 34-37 + aortic arches 7-11 = “the two number facts” — Homomeric metamerism = most segments alike (only clitellar differ) — Peristaltic LOCOMOTION = circular + longitudinal muscle + coelom — Coelomic fluid pressure = HYDROSTATIC skeleton (no bones) — Pharynx + gizzard + intestine = digestive tract (soil ingest) — Hermaphrodite but SELF-fertilisation impossible (cross-fertilisation)	OUTLET D — INTERNAL SYSTEMS OUTLETS — CIRCULATION: closed; 2-3 pairs aortic arches (seg 7-11) = “heart” dorsal + ventral vessel + setal vessels; blood = red (chlorocruorin) — EXCRETION: METANEPHRIDIA (one pair/segment; 100+ total) flame cup (nephrostome) → tubules → nephridiopore (coelom → outside) — NERVE: brain ganglion + ventral cord + giant fibre + circum-oesophageal ring Perivisceral nerves supply each segment organ — RESPIRATION: through moist body surface (diffusion); no lung/gill Dry soil kills it (gills/surface dry) → emerges after rain — SENSE: light + touch; photoreceptor in skin; eyes = none Auditory: none; but sensitive to vibration (seismic)
OUTLET E — METAMERISM + REPRODUCTION — METAMER = one segment; METAMERISM = repeated segments — Homomeric = all similar (polychaete); heteromeric = specialised — Somatic + visceral + genital segments = “double metamerism” — REPRODUCTION: hermaphrodite; two males + two female genital pores — CROSS-fertilisation; clitellum builds COCOON; eggs + sperm inside — NO larva (direct development); embryo → young earthworm — Regeneration = HEAD / TAIL regrow (polychaete: polyphyllia) — “Earthworms = soil aerators” = burrow + castings = plant food	OUTLET F — EARTHWORM vs LEECH vs POLYCHAETE — Earthworm: setae, no parapodia, soil, hermaphrodite — Leech: suckers, few setae, water, 21 fixed segments, hermaphrodite — Polychaete: parapodia + many setae, marine, separate sexes, larva — ID: “has parapodia” = polychaete; “has suckers” = leech; “setae only” = oligochaete — Leech = 2 suckers (oral + posterior); some parasitic on vertebrates — <i>Hirudo</i> = medicinal leech (historically for blood-letting) — Polychaete regeneration = polyphyllia (split worm → 2 worms) — <i>Nereis</i> = “sand worm” / “sea horse worm”; fastest polychaete swimmer
MNEMONICS + EXAM GUTS + ONE-PAGE RECAP — Class “P-O-H” = Polychaeta (parapodia, sea) · Oligochaeta (setae, soil) · Hirudinea (suckers, leech) — 2 number facts: CLITELLUM 34-37 + AORTIC ARCHES 7-11 (earthworm exam) — “3-H-T”: 3 classes · Hermaphrodite · True coelom (schizocoel + protostome) — Metamerism = “repeated cell/tissue blocks” → links back to Sec.1 cell organisation — Coelomic fluid = hydrostatic skeleton = peristaltic burrowing (no skeleton) — Reproduction: hermaphrodite BUT no self-fertilisation = cross-fertilisation + cocoon — GUT: “parapodia present” = polychaete; “suckers” = leech; “setae only” = oligochaete — GUT: leech = 21 fixed segments; polychaete = separate sexes + trochophore larva	

Figure 6: Annelida — all single outlets on one page